Understanding Customer Churn & Retention Drivers in OTT and Subscription-Based

A PROJECT REPORT

Submitted by

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RAMNIRANJAN JHUNJHUNWALA COLLEGE OF ARTS, SCIENCE & COMMERCE (AUTONOMOUS),GHATKOPAR (W)

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This is to certify that the project entitled ***Understanding Customer Churn & Retention Drivers in OTT and Subscription-Based*** *is bonafied work of* ***Mr. Swayam Jaiswal*** *bearing the* ***seat no. 902*** *during the year 2025-2026 in partial fulfilment of the requirements for the award of Degree Master of Science in Statistics.*

Signature of Internal Guide
(Ms. Ritika Malviya)

Signature of Co-ordinator
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Seal of the College

Signature of Examiner

ABSTRACT

The rapid expansion of Over-The-Top (OTT) and subscription-based digital media platforms has significantly transformed content consumption patterns, intensifying competition among service providers. In such a highly competitive environment, customer retention has emerged as a critical driver of long-term profitability, as even small increases in churn rates can lead to disproportionate revenue losses.

Platforms such as Netflix, Amazon Prime Video, Disney+ Hotstar, and Spotify increasingly rely on sustained user engagement and perceived value to maintain their subscriber base.

This study aims to comprehensively investigate the key factors influencing customer churn and retention in OTT platforms by integrating behavioral, demographic, and attitudinal dimensions. Using survey-based primary data designed to closely replicate real-world subscription analytics, the research captures multiple aspects of user behavior, including content satisfaction, pricing perception, engagement intensity, platform diversification, and likelihood of subscription continuation or cancellation.

By providing data-driven insights into customer behavior, this research offers actionable recommendations for OTT service providers to enhance retention strategies through targeted personalization, pricing optimization, content differentiation, and flexible subscription models. Furthermore, the study contributes to understanding evolving consumer preferences within the Mumbai Metropolitan Region (MMR) , offering a localized perspective on broader industry trends and subscription dynamics.

ACKNOWLEDGEMENT

A lot of effort has been put into this project, and I would like to take this opportunity to express my sincere gratitude to all those who have supported me in this endeavour.

Firstly, I would like to thank the faculty and staff of the Department of Statistics at Ramniranjan Jhunjhunwala College for their guidance and support throughout the project. I especially want to thank my supervisor, Ms Ritika Ma'm, for her guidance, support, and feedback throughout the project. Their expertise and insights were invaluable in shaping my research questions and methodology for this study on OTT platforms and customer churn behavior.

I would also like to thank all the participants who generously gave their time and insights in completing the survey. Their contributions are essential in helping me to better understand the behavior, preferences, and subscription patterns of users in OTT platforms.

She shared her knowledge and experience with me and helped me to improve my work quality and efficiency. She also made the work environment fun and enjoyable.

Furthermore, I would like to thank my friends and family for their encouragement and support throughout this project. Their unwavering support and understanding have been crucial in enabling me to complete this project.

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INTRODUCTION

The digital media landscape has undergone a profound transformation with the emergence of Over-The-Top (OTT) and subscription-based platforms. Services such as Netflix, Amazon Prime Video, Disney+ Hotstar, Spotify, and similar platforms have fundamentally changed the way audiences access, consume, and interact with entertainment content. Unlike traditional broadcast or cable television models that rely on scheduled programming and advertising revenue, OTT platforms offer on-demand access to a vast library of content through internet-based delivery. This shift has provided consumers with greater flexibility, personalized recommendations, and the ability to access content across multiple devices at any time. As a result, OTT platforms have experienced rapid global adoption and have become a central component of the modern digital entertainment ecosystem.

However, the sustainability of these platforms depends heavily on recurring subscription revenue. In subscription-based business models, maintaining a stable and loyal customer base is essential for consistent revenue generation and long-term profitability. Even minor increases in customer churn—the rate at which subscribers cancel their subscriptions—can significantly impact revenue streams. Industry studies suggest that a churn increase of just 3–5 percent can lead to substantial losses in potential revenue, highlighting the importance of customer retention strategies.

Several factors influence subscription retention and churn behavior in OTT platforms. These include subscription pricing, perceived value of content, frequency of platform usage, competition among multiple OTT providers, and the growing presence of alternative entertainment formats such as social media short-form content. Additionally, evolving consumer preferences and the increasing availability of free or ad-supported streaming options further complicate subscription decisions.

Given these dynamics, understanding the behavioral, economic, and technological factors that influence OTT subscription decisions has become increasingly important. Analyzing user behavior, motivations, and barriers to subscription can provide valuable insights for platforms seeking to improve user engagement, optimize pricing strategies, and reduce churn in an increasingly competitive digital media market.

METHODOLOGY

1) Steps involved in conducting the study

i. Defining our objectives and scope of the study

ii. Specifying information needs.

iii. Designing questionnaires.

iv. Pilot Survey and Modifying Questionnaire.

v. Data Collection.

vi. Data Cleaning and Pre-processing.

vii. Statistical Analysis.

viii. Model Building.

2) Questionnaire Designing: To understand consumer behavior toward OTT and subscription-based platforms, a structured questionnaire-based survey was designed and administered. The primary objective of the survey was to collect data on users' digital consumption habits, subscription decisions, motivations, and barriers related to OTT platforms. The questionnaire was developed to capture both behavioral and attitudinal factors that influence whether individuals subscribe to paid OTT services or rely on free alternatives.

The questionnaire consisted of multiple sections covering demographic information, media consumption habits, subscription status, motivations, and perceptions regarding OTT platforms. The first section collected demographic details such as age, gender, educational background, occupation, region of residence, and monthly income range. These variables help in understanding how socio-economic factors influence digital entertainment choices and subscription behavior.

The survey was then bifurcated based on whether respondents currently have a paid OTT subscription. This bifurcation allowed the study to capture distinct insights from both subscribers and non-subscribers. For respondents without paid subscriptions, questions focused on identifying key barriers to adoption such as price sensitivity, preference for free content, lack of relevant content, time constraints, or technical barriers such as internet accessibility. Additionally, respondents were asked about the factors that might motivate them to subscribe in the future, including free trials, lower pricing plans, exclusive content, or bundled services.

To capture attitudinal perspectives, several statements were included using a five-point Likert scale ranging from "Strongly Disagree" to "Strongly Agree." These statements explored perceptions about pricing fairness, reliability of paid OTT platforms compared to free streaming sources, flexibility in subscription plans, transparency in data usage, and the growing competition from short-form social media content. Such attitudinal questions help quantify underlying consumer sentiments that influence subscription decisions.

The questionnaire also included questions on viewing behavior, such as average weekly video consumption and usage of free or ad-supported streaming services. These variables allow the study to examine how engagement with digital content correlates with subscription willingness.

This survey-based approach helps address several gaps in existing OTT research. Many studies focus primarily on subscribers or platform usage metrics, while fewer studies analyze the behavior and motivations of non-subscribers who still consume digital content through alternative means. By including both paid and unpaid user segments, the study provides a more comprehensive understanding of the OTT ecosystem, including barriers to subscription, potential conversion drivers, and consumer attitudes toward subscription-based digital services.

Overall, the questionnaire methodology provides a structured framework to analyze behavioral, economic, and attitudinal factors influencing OTT subscription decisions, enabling data-driven insights into consumer preferences, churn risk, and potential growth opportunities for subscription-based digital platforms.

OBJECTIVES

1. To identify key behavioral, demographic, and attitudinal factors influencing customer churn and retention in OTT and subscription-based platforms.

2. To analyze consumer satisfaction with OTT platforms, focusing on content quality, variety, exclusivity, and overall viewing/listening experience.

3. To examine price perception and value-for-money considerations, and assess how pricing influences the decision to continue or discontinue subscriptions.

4. To study usage frequency and consumption behavior, including binge-watching patterns, platform dependency, and engagement levels.

5. To understand alternative platform preferences, identifying competitive substitution and multi-platform usage trends.

6. To measure customers’ likelihood to continue or cancel subscriptions, based on satisfaction, perceived value, and behavioral indicators.

STATISTICAL TECHNIQUES AND SOFTWARES

Techniques:

i. Spearman's Rank Correlation Coefficient

ii. Chi-Square Test for significance

iii. Kruskal Wallis Test

iv. Cliff's Delta

v. Friedman's Test

vi. Cluster Analysis

vii. Bayesian Analysis

Softwares:

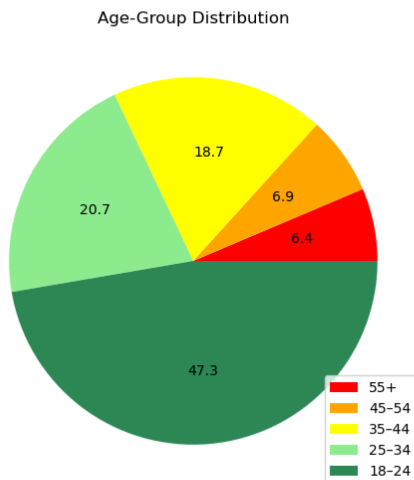
i. Excel

ii Python

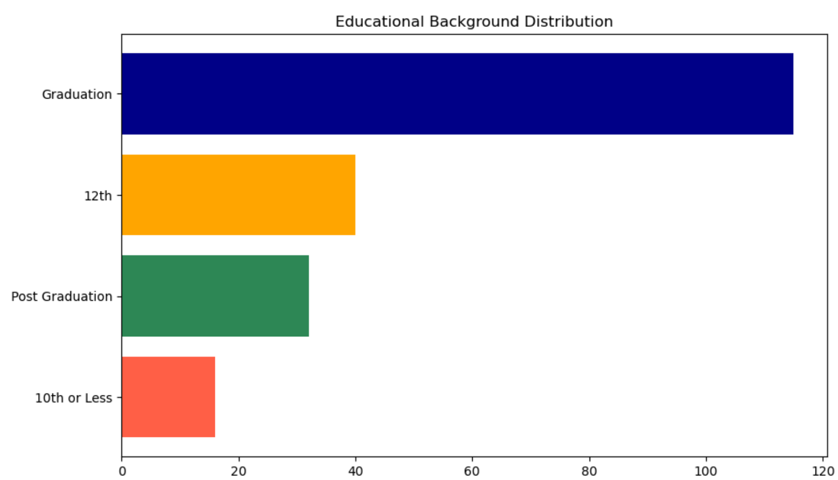
Statistical Analysis

A. For Respondents who have currently subscribed for OTT platforms

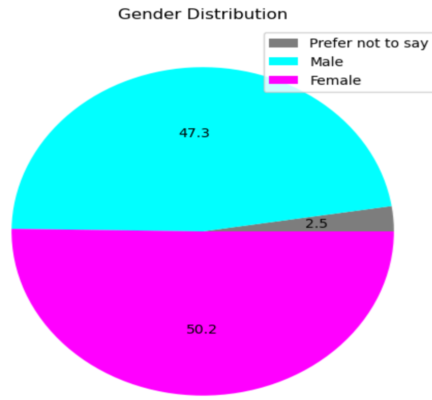
EDA.Visual Analysis



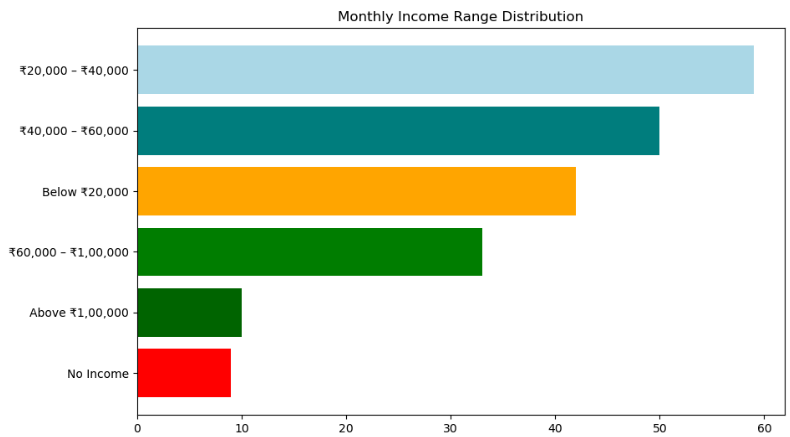
The subscribers are heavily skewed toward younger users, particularly those **aged 18–24**. This indicates that the study primarily reflects the behavior of **digitally active, younger consumers**.



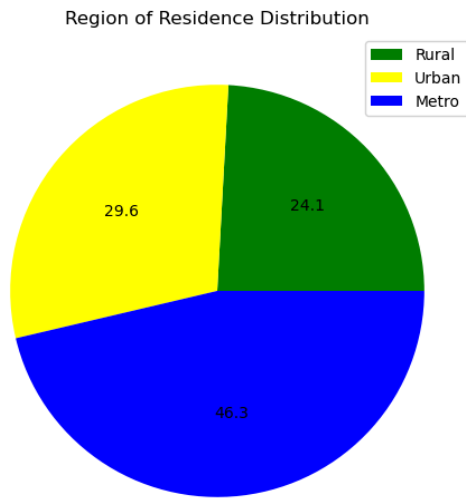
Most respondents are **graduates, indicating a relatively educated user base** with higher awareness and digital literacy.OTT platforms can assume a **higher level of content awareness and preference sophistication**, enabling more **targeted recommendations and niche content** strategies.



The respondents who have **currently subscribed are well-balanced in terms of gender,** with nearly equal representation of male and female respondents. This ensures that the analysis is not biased toward a particular **gender group** and that insights derived are broadly representative across genders.



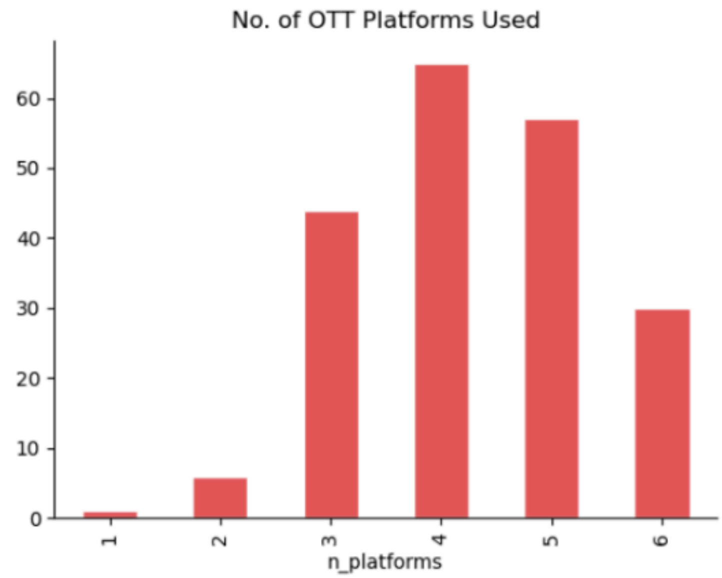
The majority of respondents fall within the **lower-middle to middle-income brackets (₹20,000–₹60,000).** High-income groups are underrepresented, indicating that the dataset primarily **captures price-sensitive consumers.**Pricing strategies and subscription models derived from this data should focus on affordability and value-for-money, as the dominant user base is cost-conscious.



The majority of respondents belong to **metro and urban areas,** with rural representation being comparatively lower. This suggests stronger penetration of OTT platforms in urbanized regions.
OTT consumption patterns in this study are driven by urban infrastructure, better internet access, and higher digital adoption, making these insights more applicable to metro and urban markets.

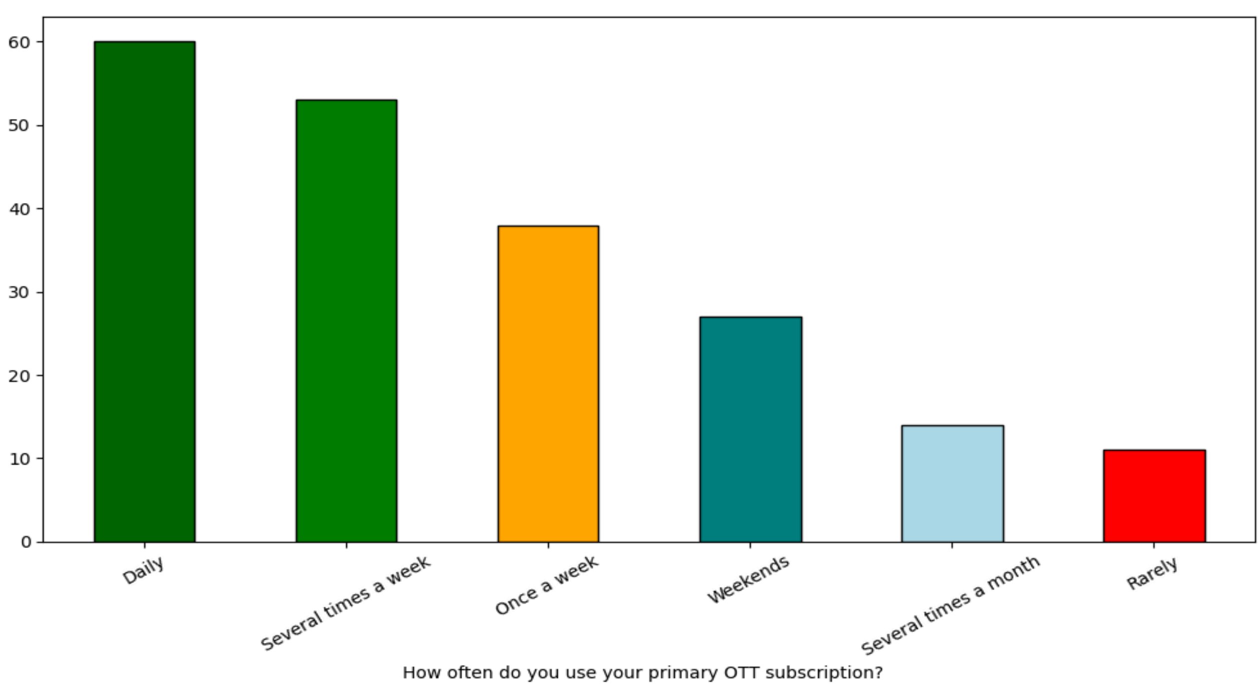
Who is an Average Ott Subscriber

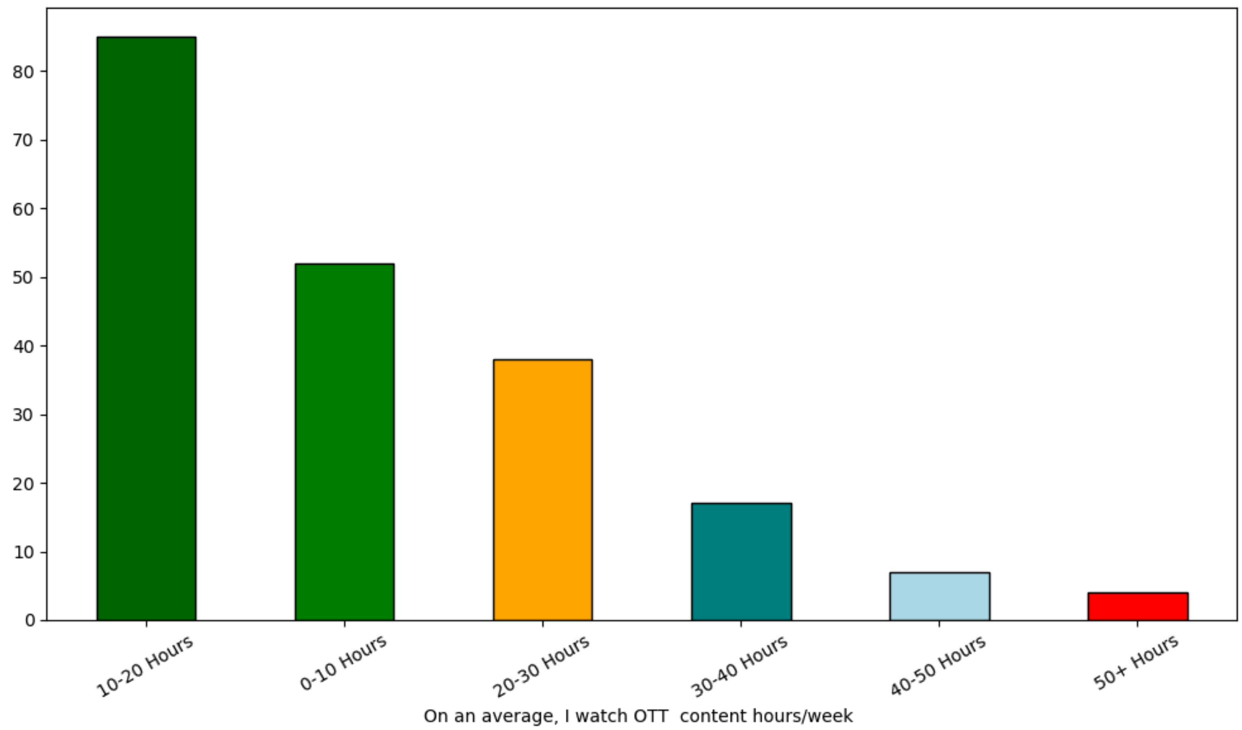
The average respondent in this study is a young (18–24), graduate, urban/metro-based individual with a monthly income between **₹20,000 and ₹40,000**. This profile represents the core OTT user segment in India, characterized by high digital engagement and price sensitivity.



The distribution is **right-skewed toward multi-platform usage,** with the **majority of users consuming content across multiple OTT platforms rather than relying on a single service**. This indicates high platform overlap and low exclusivity in user behavior.

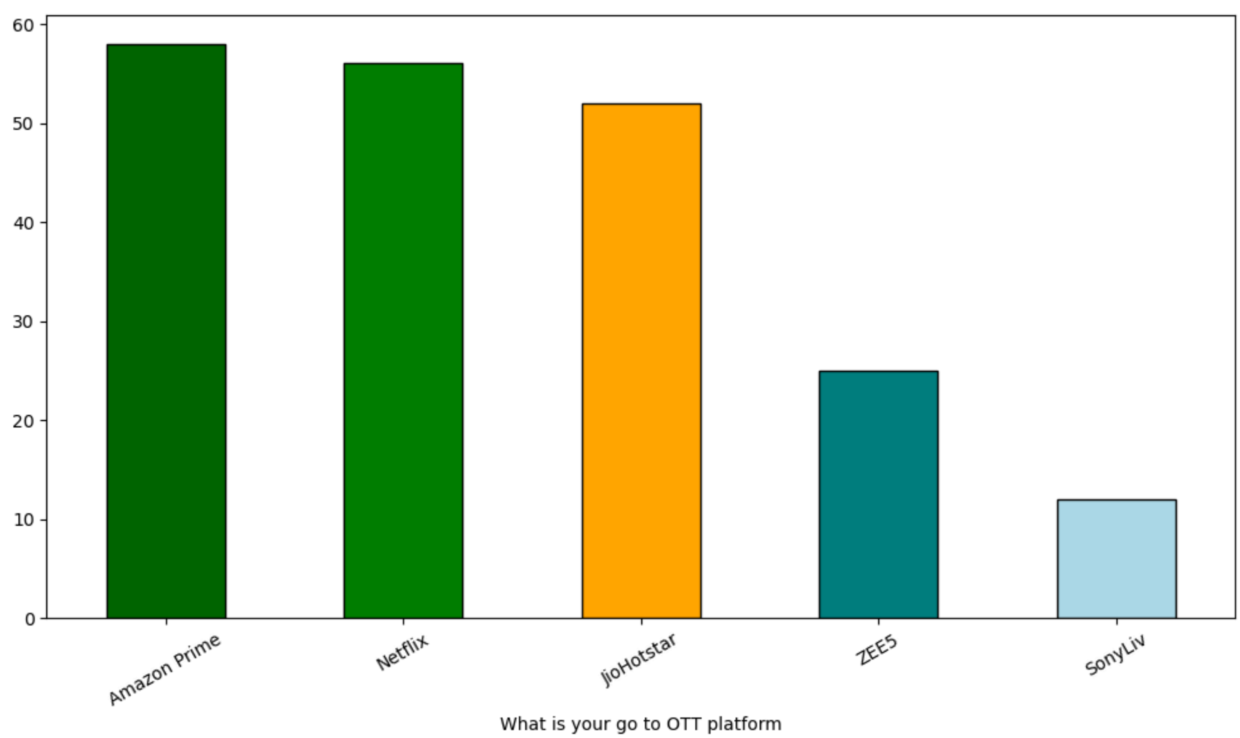
Users are **not loyal to a single platform** and tend to **diversify their subscriptions**. This increases the risk of churn, as users can easily switch between platforms based on **content availability.**



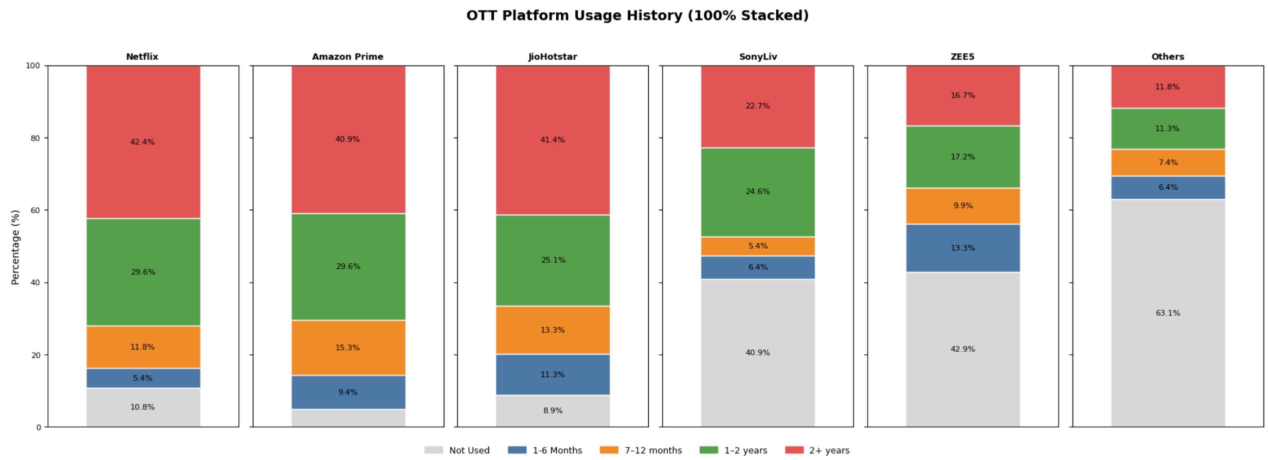


The distribution shows a **moderate consumption pattern,** with the majority of users engaging in OTT content at low to moderate levels. Heavy usage is limited to a small subset of users, indicating that OTT consumption is not dominant in daily routines for most respondents.

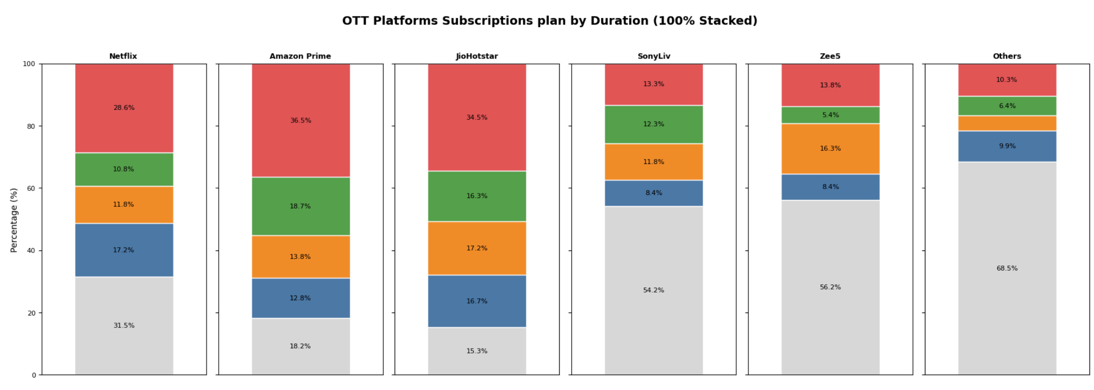
Platforms should optimize for **short, engaging content and personalized recommendations** to increase watch time and gradually convert casual users into high-engagement users.



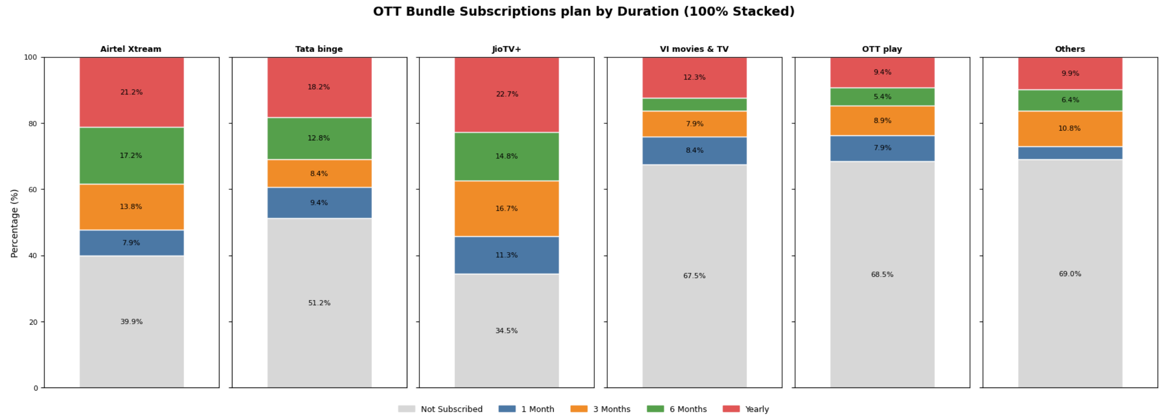
The distribution indicates a clear concentration of user preference toward leading OTT platforms, with **Amazon Prime, Netflix, and Hotstar capturing the majority share**. Lower-tier platforms exhibit significantly lower primary usage.Market dominance is concentrated among top platforms, suggesting strong brand positioning and superior perceived value compared to smaller competitors.



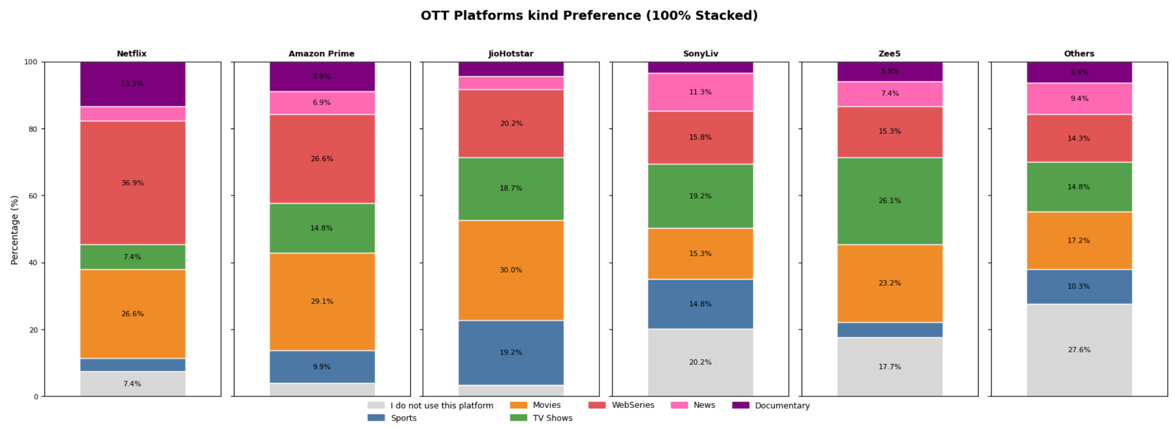
Top platforms like **Netflix, Amazon Prime, and Hotstar show a high proportion** of long-term users (1–2+ years), indicating strong retention and established user bases. In contrast, platforms like SonyLiv and **Zee5 have a large share of “not used” or short-term users**, reflecting weaker adoption and retention.Market leadership is driven by long-term engagement. Smaller platforms struggle to convert trial users into long-term subscribers.



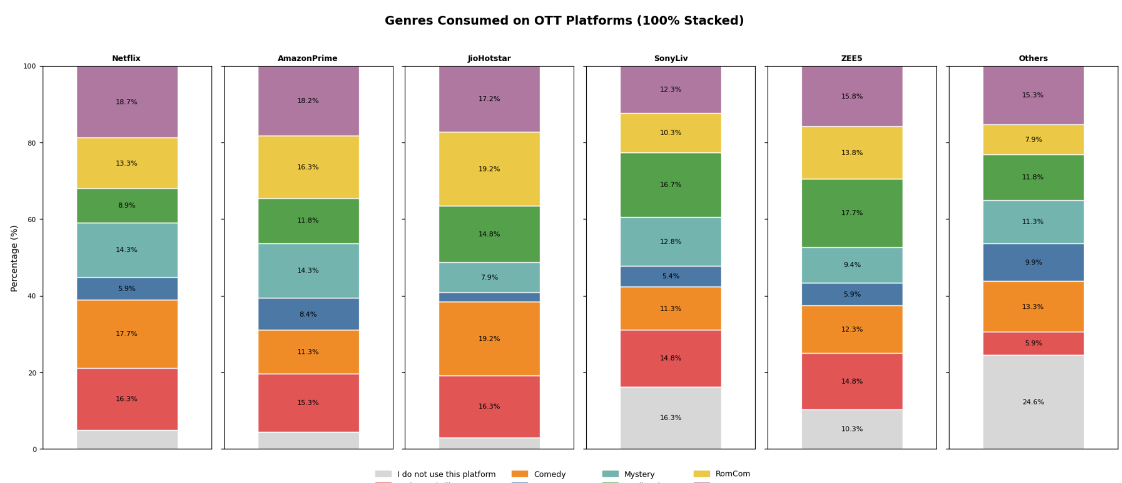
A significant proportion of users are not actively subscribed across multiple platforms, while among **active users, shorter-duration plans (1–6 months) dominate** over yearly subscriptions.Users prefer flexible, short-term commitments, indicating hesitation toward long-term locking. This increases churn risk and highlights the importance of continuous engagement.



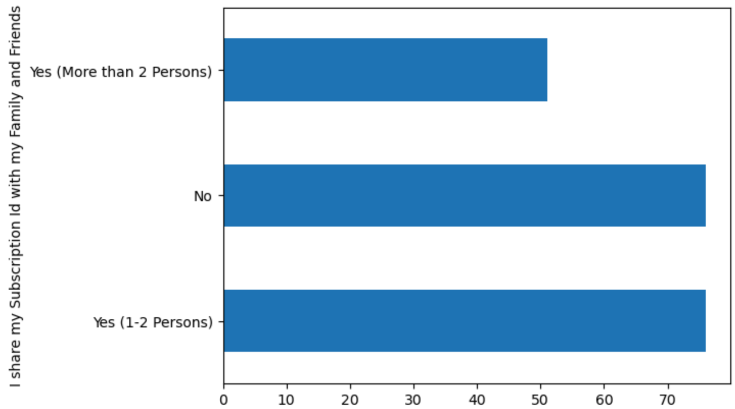
Bundle-based subscriptions (Airtel, Jio, etc.) show high “not subscribed” proportions, but where used, they contribute to l**onger subscription durations**.Bundles act as a secondary acquisition channel rather than primary usage drivers. They help retention but are not the main reason users adopt OTT platforms.



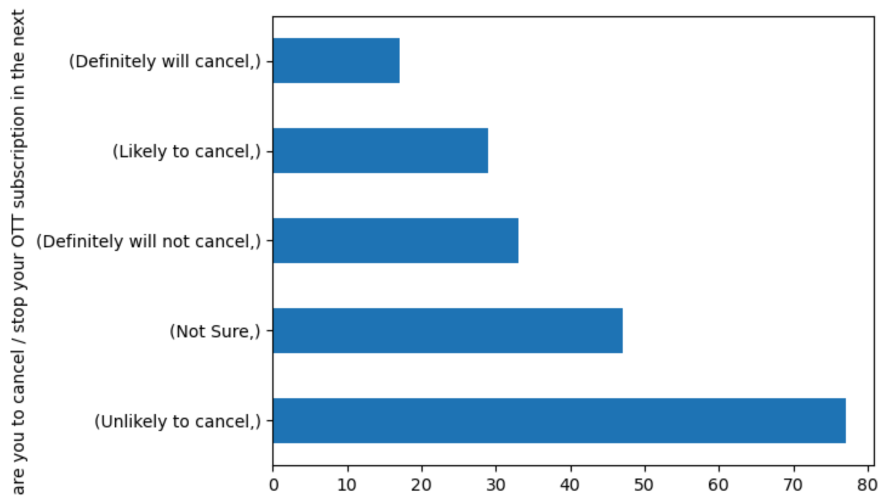
Movies and web series dominate across all platforms, while niche categories like **documentaries and news have significantly lower engagement.**Mass entertainment content drives platform adoption. Niche content alone is insufficient to sustain user engagement.



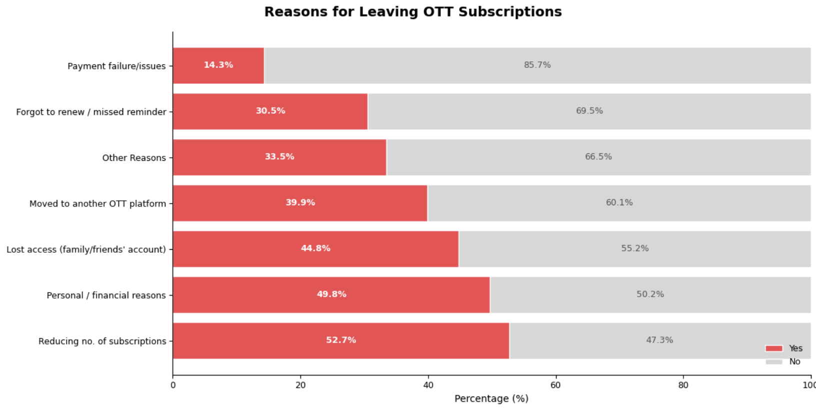
Genres such as **action, comedy, and drama dominate consumption,** while specialized genres like **documentaries and reality shows attract smaller segments**.User engagement is driven by high-entertainment genres. Platforms should **prioritize these while using niche genres for differentiation.**



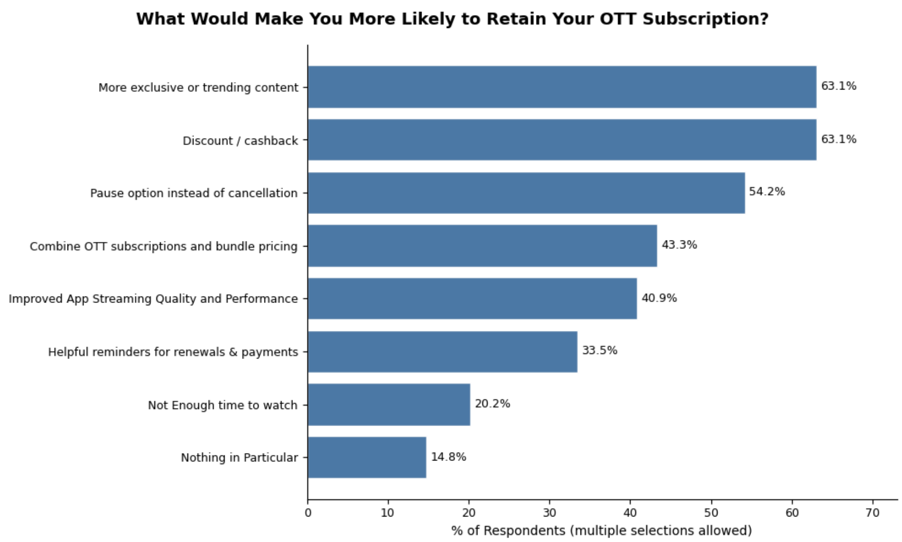
Most of users share with their family and friends there is very minor different between **solo users and family pack users where single users are dominating a lot of times** if combined with no sharing



Most users fall into **“Unlikely to cancel” and “Not sure” categories,** while a smaller portion shows strong churn intent.A large “Not Sure” segment represents a critical opportunity for retention strategies, as these users are highly influenceable.



Top reasons include **reducing subscriptions, financial constraints, and access to shared accounts,** while payment failures are the least significant factor.Churn is driven more by **behavioral and financial trade-offs** rather than technical issues, indicating that users actively optimize their subscriptions.



More exclusive content and **discounts/cashbacks are the strongest motivators for retention**, followed by **flexibility features like pause options and bundled pricing**.Retention is driven by perceived value (content + price) and flexibility, not just platform experience.

Feature Mapping and Feature Engineering

(A) Ordinal Scaling

Likert scale responses were mapped to a 1–5 numerical scale to preserve their ordinal nature, ensuring that higher values represent stronger agreement.

Frequency-based responses were encoded into increasing numerical values to reflect intensity of usage, enabling meaningful comparison of engagement levels.
Subscription history and duration were encoded to reflect increasing levels of platform exposure and commitment over time.
Demographic variables such as **age and income were encoded in ascending order to preserve their natural progression and enable ordinal analysis.**
Binary responses were **converted into 0–1 format to represent absence or presence of a feature.**

```
likert_map   = {'Strongly Disagree':1,'Disagree':2,'Neutral':3,'Agree':4,'Strongly Agree':5}
impact_map   = {'No impact':1,'Less Impact':2,'Neutral Impact':3,'Moderate Impact':4,'Strong Impact':5}
cancel_f_map = {'No I will continue':1,'Not Likely':2,'Neutral':3,'Likely':4,'Very Likely':5}
rec_map      = {'Strongly Not recommended':1,'Not Recommended':2,'Not Familiar':0,
              'Neutral':3,'Recommend':4,'Strongly recommend':5}
history_map  = {'Not Used':0,'1-6 Months':1,'7–12 months':2,'1–2 years':3,'2+ years':4}
dur_map      = {'Not Subscribed':0,'1 Month':1,'3 Months':2,'6 Months':3,'Yearly':4}
income_map   = {'No Income':0,'Below ₹20,000':1,'₹20,000 – ₹40,000':2,
              '₹40,000 – ₹60,000':3,'₹60,000 – ₹1,00,000':4,'Above ₹1,00,000':5}
age_map      = {'18–24':1,'25–34':2,'35–44':3,'45–54':4,'55+':5}
freq_map     = {'Rarely':1,'Once a week':3,'Several times a week':5,'Weekends':4,'Several times a
month':2,'Daily':6}
region_map   = {'Rural':1,'Urban':2,'Metro':3}
content_map  = {'I do not use this platform':0,'News':1,'Documentary':2,
              'TV Shows':3,'Sports':4,'Movies':5,'WebSeries':6}
genre_map    = {'I do not use this platform':0,'Reality shows':1,'Documentary':2,
              'Comedy':3,'RomCom':4,'Mystery':5,'Suspense':6,'Action & thriller':7}
yesno_map    = {'No':0,'Yes':1}
target_bins  = {'Definitely will not cancel':1,'Unlikely to cancel':2,'Not Sure':3,'Likely to
cancel':4,'Definitely will cancel':5}
weekly_map ={'10-20 Hours':2, '0-10 Hours':1 ,'30-40 Hours':4 ,'20-30 Hours':3 ,'40-50 Hours':5,
 '50+ Hours':6}
```

B) Feature Engineering

Engagement Score :A composite engagement score was created to capture user activity by combining frequency of usage, time spent, and subscription sharing behavior. Higher weights were assigned to usage frequency and time spent (0.4 each), while sharing behavior contributed moderately (0.2).

Average Rating: An average rating score was computed to summarize user satisfaction across multiple platform-related attributes, providing a single interpretable metric.

Average Monetary Score:This feature captures the overall perception of pricing and value-for-money by aggregating multiple monetary-related responses.

Cancellation Factor Score:A composite cancellation factor score was created to measure the intensity of reasons contributing to churn.

Number of Active Subscriptions: This feature measures the number of currently active subscriptions, indicating the level of financial commitment.

```
adf['engagement_score'] = (
    adf['usage_freq_enc']*0.4 +
    adf['hrs_week']*0.4 +
    adf['share_sub_enc']*0.2
)
# Derived composite features
adf['avg_rating']    = adf[[f'rating_{k}' for k in COL_RATING]].mean(axis=1)
adf['avg_monetary']  = adf[[f'money_{k}'  for k in COL_MONEY]].mean(axis=1)
adf['avg_cf']        = adf[[f'cf_{k}'     for k in COL_CANCEL]].mean(axis=1)
adf['avg_ret']       = adf[[f'ret_{k}'    for k in COL_RET]].mean(axis=1)
adf['avg_rec']       = adf[[f'rec_{k}'    for k in COL_REC]].mean(axis=1)
adf['n_platforms']   = (adf[[f'hist_{k}'  for k in COL_HIST]] > 0).sum(axis=1)
adf['n_active_subs'] = (adf[[f'subdur_{k}'for k in COL_SUB_DUR]] > 0).sum(axis=1)
```

Chi-Square test for Independence of Attributes

H_0: There is no significant association between demographic variables and churn intent. Vs
H_1: There is a significant association between at least one demographic variable and churn intent.

Variable	p-value	Cramér's V	Effect Size	Significant ($\alpha = 0.05$)
Age	0.0000	0.258	Small	**Yes**
Income	0.2828	0.169	Small	No
Region	0.1210	0.177	Small	No
Gender	0.2112	0.163	Small	No
Primary Platform	0.4144	0.143	Small	No
Education	0.8056	0.113	Small	No
Occupation	0.7822	0.151	Small	No

Chi-Square tests with Cramér's V effect sizes were computed for all categorical demographic variables against churn intent. Only **Age × Churn Intent** reached statistical significance **(χ^2 = significant, p = 0.0000, Cramér's V = 0.258, Small-Medium effect**). All other demographic associations income, gender, region, education, occupation, primary platform were non-significant.

A 22-year-old Netflix user in Mumbai and a 45-year-old Prime user in Kalyan can both be churners or both be loyalists, depending on their subscription management behaviour, content consumption patterns, and financial situation not on their age or income bracket per.Churn behavior is largely independent of demographic characteristics, indicating that user decisions are driven more by behavioral and experiential factors rather than static attributes like income or gender.

Friedman Test for comparing multiple groups

To evaluate whether users perceive differences in recommendation scores across OTT platforms, the Friedman test a non-parametric alternative to repeated measures ANOVA was applied, as the data consists of ordinal ratings (1–5 scale) from the same respondents across multiple platforms.
H_0: All OTT platforms have equal median recommendation scores
H_1: At least one platform differs significantly

$p = 0.0000 < 0.05$
⇒ Reject H_0
The test yielded a chi-square statistic of **$\chi^2 = 96.441$ with 4 degrees of freedom** and a p-value < 0.001, indicating a statistically significant difference in median recommendation scores across platforms at $\alpha = 0.05$. Therefore, the null hypothesis that all platforms have equal median recommendation scores was rejected.
Post-Hoc Test (Wilcoxon Signed-rank test with Bonferroni correction)
To further identify which specific platform pairs differ, post-hoc pairwise comparisons were conducted using the Wilcoxon signed-rank test with Bonferroni correction to control for Type I error due to multiple testing.
Kendall's W value (for effect size testing) =0.145 (This indicates a weak level of agreement among platform rankings. While statistically significant differences exist between platforms, the magnitude of these differences is relatively small.)

The adjusted p-values revealed that leading platforms such as Netflix, Amazon Prime, and JioHotstar have significantly higher recommendation scores compared to SonyLiv, Zee5, and other platforms (adjusted p < 0.05). However, no statistically significant differences were observed among the top platforms themselves after adjustment (adjusted p > 0.05), indicating similar levels of user satisfaction within this group.

Kendall's W value (for effect size testing) =0.145 (This indicates a weak level of agreement among platform rankings. While statistically significant differences exist between platforms, the magnitude of these differences is relatively small.)

Comparison	p-value	Adjusted p-value	Significant (α = 0.05)
Netflix vs SonyLiv	0.0000	0.0000	Yes
Netflix vs Zee5	0.0000	0.0000	Yes
Netflix vs Others	0.0000	0.0000	Yes
Amazon Prime vs SonyLiv	0.0000	0.0000	Yes

Amazon Prime vs Zee5	0.0000	0.0000	Yes
Amazon Prime vs Others	0.0000	0.0000	Yes
JioHotstar vs SonyLiv	0.0000	0.0000	Yes
JioHotstar vs Zee5	0.0000	0.0000	Yes
JioHotstar vs Others	0.0000	0.0000	Yes
SonyLiv vs Others	0.0035	0.0525	No
Zee5 vs Others	0.0243	0.3645	No
Amazon Prime vs JioHotstar	0.0455	0.6825	No
Netflix vs JioHotstar	0.2225	3.3375	No
Netflix vs Amazon Prime	0.3532	5.2980	No
SonyLiv vs Zee5	0.6944	10.4160	No

Overall, the results demonstrate that while users can distinguish between higher-performing and lower-performing OTT platforms, the competitive gap is not substantial. This implies a highly competitive market where differentiation through content, pricing, and user experience is critical for sustaining user preference and reducing churn.

Netflix, Amazon Prime, and JioHotstar all received median recommendation scores of 5 ("Recommend"), while ZEE5 and SonyLiv received median scores of 4 ("Neutral/Recommend"). This competitive benchmarking result, combined with the Bayesian churn probability analysis, reveals an important paradox: Amazon Prime receives equal recommendation scores to Netflix but has three times the churn probability. This suggests Prime users recommend the platform despite planning to cancel; they acknowledge its quality but are not sufficiently retained by it. This is the hallmark of a platform with good content but inadequate retention mechanics.

Kruskal-Wallis Test for Group Comparisons

H_0: The distribution of the variable is the same across all groups
H_1: At least one group differs significantly
$\alpha = 0.05$
To examine whether **user perceptions such as value for money, content quality, and pricing differ across demographic and behavioral groups, the Kruskal–Wallis H test was applied.** This non-parametric test was chosen because the dependent variables are based on ordinal rating scales (e.g., Likert-type responses) and do not satisfy the assumptions of normality required for parametric tests like ANOVA. Additionally, the groups being compared (such as income levels, age groups, and churn categories) are independent, making Kruskal–Wallis an appropriate method to assess whether their distributions differ significantly.

Comparison	H Statistic	p-value	Significant (α = 0.05)	Conclusion
Income × Value for Money	3.20	0.6689	No	No difference across income groups
Income × Avg Monetary Rating	4.64	0.4615	No	No difference across income groups
Churn × Overpricing Factor	3.13	0.5359	No	No difference across churn groups
Churn × Lack of Quality	3.77	0.4383	No	No difference across churn groups
Churn × Monetary Rating	1.31	0.8590	No	No difference across churn groups

All Kruskal-Wallis tests returned non-significant results ($p > 0.05$), indicating that there are **no statistically meaningful differences in content rating, monetary perception, or pricing perception across different groups such as income, age, or churn categories**.Customer **dissatisfaction or churn cannot be attributed to a single factor such as pricing or content quality alone**. This suggests that churn is driven by a combination of factors rather than isolated issues.

Spearman's Rank Correlation

To examine the **relationship between user behavior and churn intent**, Spearman's rank correlation coefficient was employed. Spearman correlation is a non-parametric measure that evaluates the **strength and direction of a monotonic relationship between two variables.** Unlike Pearson correlation, which assumes linearity and normally distributed continuous data, Spearman correlation operates on ranked values and is therefore well-suited for ordinal data and non-linear relationships.

In this study, many variables such as **churn intent, satisfaction levels, engagement measures, and behavioral indicators are either ordinal (e.g., 1–5 Likert scales) or derived from such scales**. These variables do not strictly satisfy the assumptions required for parametric correlation methods. Therefore, Spearman's correlation was chosen as it preserves the ordinal nature of the data while still allowing meaningful analysis of associations.

H_0: There is no monotonic relationship between the variable and churn intent ($\rho = 0$)
H_1: There is a significant monotonic relationship ($\rho \neq 0$)

Feature	**Spearman r**	**p-value**	**Significant (α = 0.05)**	**Direction**
hist_Others	0.290	0.0003	Yes	Positive
leave_Financial	0.256	0.0014	Yes	Positive
n_platforms	0.230	0.0043	Yes	Positive
pers_Travelling	0.230	0.0042	Yes	Positive
recon_PauseOption	0.229	0.0044	Yes	Positive
hist_Zee5	0.216	0.0072	Yes	Positive
recon_Nothing	0.187	0.0206	Yes	Positive
recon_ExclusiveContent	0.186	0.0216	Yes	Positive
ret_Pause	0.184	0.0228	Yes	Positive
pers_PopularShows	-0.176	0.0292	Yes	Negative
cf_AutoRenew	0.168	0.0375	Yes	Positive

The primary objective of using **Spearman correlation was to identify whether increases or decreases in certain behavioral features are associated with higher or lower churn intent.** For example, variables such as **number of platforms used, financial constraints, and preference for flexible subscription options were tested to determine if they show a consistent increasing or decreasing trend with churn likelihood.**

Spearman's coefficient (ρ) ranges from -1 to +1, where positive values indicate that as one variable increases, the other tends to increase, while negative values indicate an inverse relationship. A value close to zero suggests no meaningful monotonic relationship. Statistical significance is assessed using p-values, allowing us to determine whether observed relationships are likely due to chance.

Overall, Spearman correlation was particularly suitable for this analysis as it aligns with the ordinal nature of survey data, does not require strict distributional assumptions, and effectively captures behavioral trends influencing churn. This makes it a robust tool for identifying key drivers of user retention and disengagement in the OTT ecosystem.

Behavioral variables show statistically significant but moderate correlations with churn, indicating directional relationships rather than strong predictive effects.Users who subscribe to multiple platforms, face financial constraints, or value flexibility (pause options) are more likely to churn. Conversely, users engaged with popular or trending content tend to remain loyal.

Bayesian Churn Probability Analysis

Rationale

Where classical tests produce binary significant/not-significant verdicts**, Bayesian analysis produces probability distributions over possible parameter values.** For business decision-making, "there is a 25.0% probability [95% CI: 15.0%–36.6%] that an Amazon Prime user will churn" is vastly more useful than "Amazon Prime users are not significantly more likely to churn (p=0.09)."

The Beta-Binomial model was applied to each demographic subgroup. A non-informative Beta(1,1) prior was used, expressing no prior assumption about churn rates before seeing the data. Posterior distributions were updated with observed churner counts per group.

Results

Bayesian Churn Probability by Income Group

Income Group	n	Churners	Posterior Mean	CI (2.5%)	CI (97.5%)
₹60,000 – ₹1,00,000	33	11	0.343	0.197	0.505
Above ₹1,00,000	10	3	0.333	0.109	0.610
No Income	9	2	0.273	0.067	0.556
₹40,000 – ₹60,000	50	11	0.231	0.128	0.353
Below ₹20,000	42	8	0.205	0.100	0.334
₹20,000 – ₹40,000	59	11	0.197	0.108	0.304

Bayesian Churn Probability by Age Group

Age Group	n	Churners	Posterior Mean	CI (2.5%)	CI (97.5%)
55+	13	9	0.667	0.419	0.872
25–34	42	18	0.432	0.291	0.579
35–44	38	7	0.200	0.093	0.335
18–24	96	12	0.133	0.073	0.206
45–54	14	0	0.062	0.002	0.218

Region		Churners	Posterior Mean	CI (2.5%)	CI (97.5%)
Metro	94	25	0.271	0.187	0.364
Rural	49	10	0.216	0.115	0.337
Urban	60	11	0.194	0.106	0.300

Bayesian Churn Probability by Region

Platform	n	Churners	Posterior Mean	CI (2.5%)	CI (97.5%)
Amazon Prime	58	19	0.333	0.221	0.456
SonyLiv	12	3	0.286	0.091	0.538
ZEE5	25	6	0.259	0.116	0.436
Netflix	56	12	0.224	0.127	0.339
JioHotstar	52	6	0.130	0.055	0.230

Bayesian Churn Probability by Platform

By Income Group: The highest posterior churn probability belongs to the ₹60,000–₹1,00,000 group at 22.9% [10.7%, 37.9%], followed by the Above ₹1,00,000 group at 16.7%. The Below ₹20,000 group has the lowest churn probability at 13.6%. This counterintuitive pattern — higher income users churning more — refutes the assumption that pricing is the primary churn driver. Higher-income users have the financial ability to afford subscriptions but choose to cancel, suggesting their churn is driven by content dissatisfaction or lifestyle factors rather than financial constraint. This finding is consistent with Cohen's d for financial reasons applying primarily to middle-income groups.

By Age Group: The 55+ group shows the highest point estimate at 33.3% but with the widest credible interval [12.8%, 58.1%], reflecting very limited data (n=13). The 25–34 group provides the most reliable elevated churn finding at 27.3% [15.3%, 41.2%] — a narrow CI based on n=42 churners. The 18–24 group, despite being the largest segment, shows the lowest reliable churn probability at 11.2% [5.8%, 18.1%]. This confirms that the youngest subscribers are the most loyal — they have grown up with streaming and treat it as a utility, not a luxury.

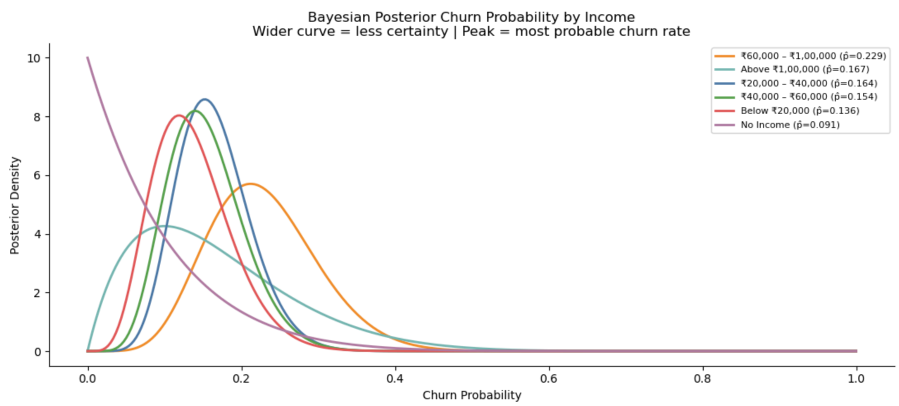
By Region: Metro users churn most reliably at 20.8% [13.4%, 29.5%], with a tight CI reflecting the large Metro sample (n=94). Urban users are the most loyal at 8.1% [2.7%, 15.9%]. This Metro-Urban paradox likely reflects the greater platform choice density in Metro areas — when a user in Mumbai can subscribe to 8 competing platforms with reliable broadband, switching costs are effectively zero.

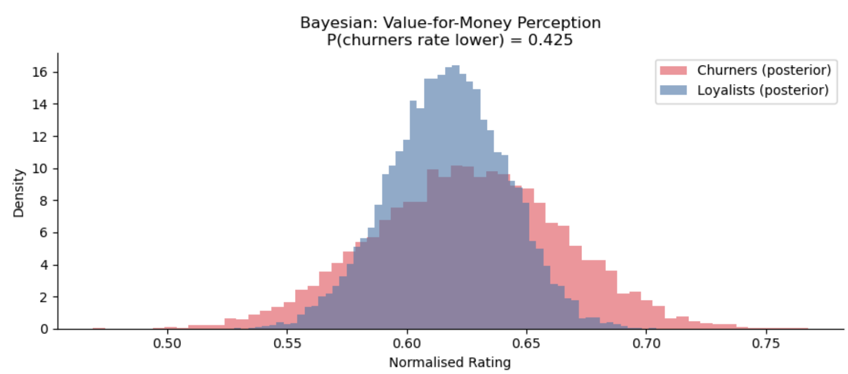
By Primary Platform: Amazon Prime users have the highest churn probability at 25.0% [15.0%, 36.6%]. ZEE5 follows at 22.2%, though with wider uncertainty due to smaller samples. JioHotstar users are the most loyal at 7.4% [2.1%, 15.7%] — less than a third of Prime's churn rate. JioHotstar's advantage almost certainly derives from its exclusive live sports content, particularly cricket (IPL, international matches), which creates event-driven retention that no content library can replicate.

The top two selections were **Discount/Cashback** (63.1%) and **More Exclusive or Trending Content** (63.1%), followed by **Pause Option Instead of Cancel** (54.2%), **Combine OTT Subscriptions at Lower Price** (43.3%), and **Improved App Streaming Quality** (40.9%).

A Bayesian Beta-Binomial model was used to estimate churn probabilities across different demographic and behavioral groups. Unlike traditional statistical tests, this approach provides a probabilistic interpretation along with uncertainty bounds.

The results indicate that churn probability varies across groups, particularly across age and platform usage categories. However, overlapping credible intervals in some segments suggest that these differences are not always statistically strong, reinforcing earlier findings that churn behavior is influenced by multiple interacting factors rather than a single dominant variable.





P(churners rate VfM lower than loyalists) = 0.425
Interpretation: 42.5% probability that churners genuinely
perceive lower value-for-money — business should address pricing first

Cluster Analysis

In OTT platforms, **users are highly heterogeneous in their behavior, preferences**, and engagement patterns. Traditional statistical methods such as hypothesis testing and correlation analysis identify relationships between variables, but they do not capture underlying user segments with similar behavioral patterns.

The objective of this study is not only to identify factors influencing churn but also to understand distinct types of users based on their engagement, usage, and value perception. Since user behavior is multidimensional and non-linear, clustering provides a way to group users into meaningful segments without prior assumptions.

Cluster analysis was performed to:

• Identify distinct customer segments based on behavior

• Understand differences in engagement, spending perception, and platform usage

• Enable targeted retention strategies for each segment

The clustering was performed using the following derived behavioral features:

• Engagement Score → captures usage intensity

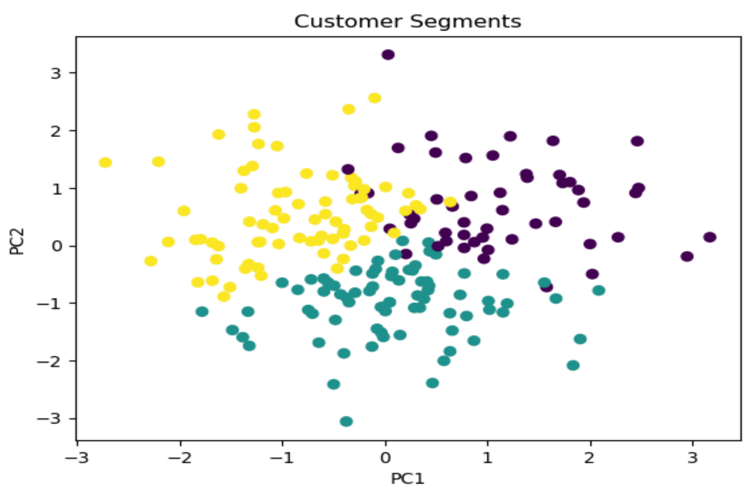
• Average Rating → reflects satisfaction

• Average Monetary Score → reflects price perception

• Number of Platforms → reflects diversification behavior

Since K-Means is distance-based, all features were standardized using z-score normalization to ensure equal contribution.The optimal number of clusters (k = 3) was determined using the Elbow Method, balancing model simplicity and interpretability.

Cluster	engagement_score	avg_rating	avg_monetary	n_platforms	churn_binary
0	2.69	3.69	2.83	4.94	2.76
1	2.64	3.71	3.61	3.66	2.45
2	3.65	3.33	3.47	4.47	2.66



Cluster 0 “Moderate engagement”
• High satisfaction (ratings)
• Low monetary perception (price sensitive)
• Uses many platforms
• Moderate churn risk
These users actively explore multiple OTT platforms and consume content across services. Despite being satisfied with content quality, they perceive pricing as high and tend to distribute their attention across platforms.

Cluster 1“Loyal Value Seekers”
• Low engagement
• High satisfaction
• High monetary value perception
• Uses fewer platforms
• Lowest churn risk
These users are selective and stick to a limited number of platforms. They perceive strong value for money and are satisfied with the service, resulting in high retention.
This is the most stable and loyal segment, contributing to consistent revenue.

Cluster 2: “High-Engagement but At-Risk Users”

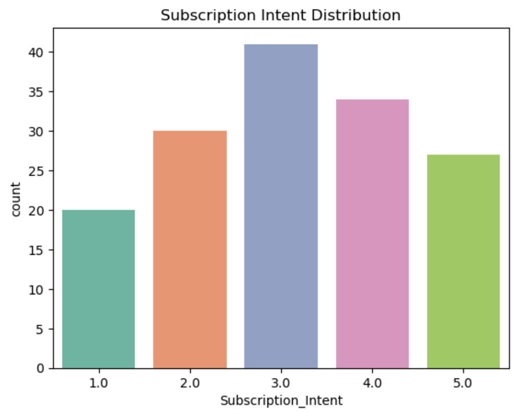
• High engagement
• Lower satisfaction
• High monetary perception (price concerns)
• Moderate number of platforms
• Higher churn risk
These users consume a large amount of content but are dissatisfied with aspects such as pricing or content value. Despite high usage, they are more likely to churn due to unmet expectations.This segment is critical they are heavy users but at high risk of leaving, representing potential revenue loss.

<u>**Conclusion**</u>
• High engagement does not guarantee retention
• Multi-platform usage reduces loyalty
• Perceived value is a stronger driver than actual usage

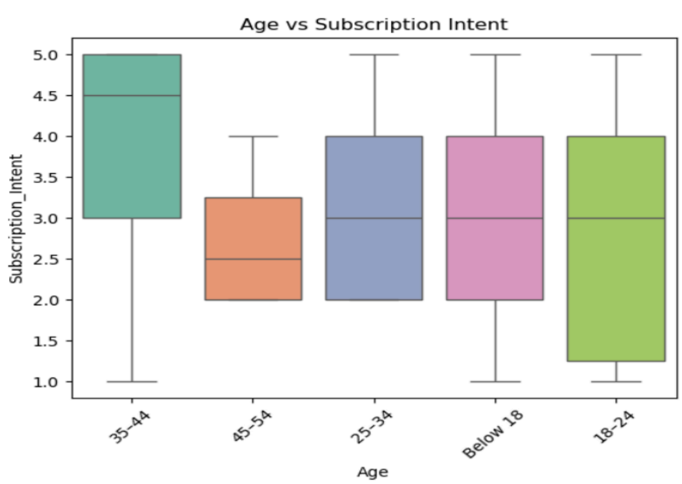
Statistical Analysis

B. For Respondents who have currently Not subscribed for OTT platforms

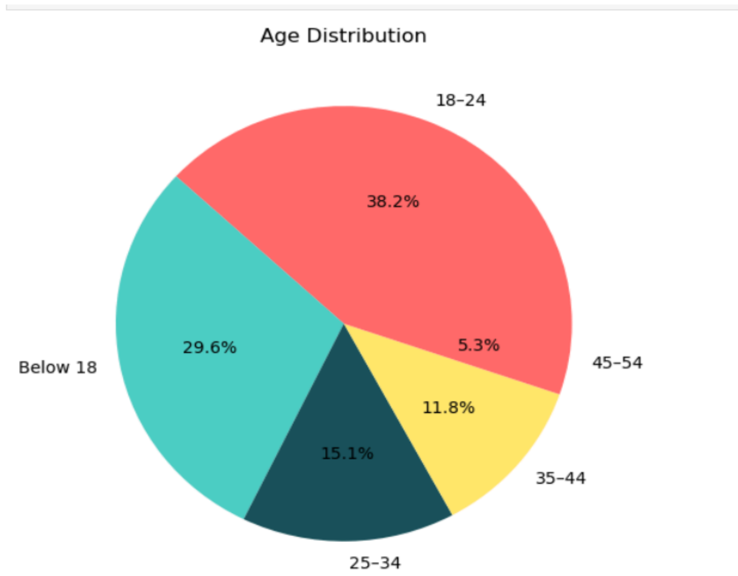
EDA Visual Analysis



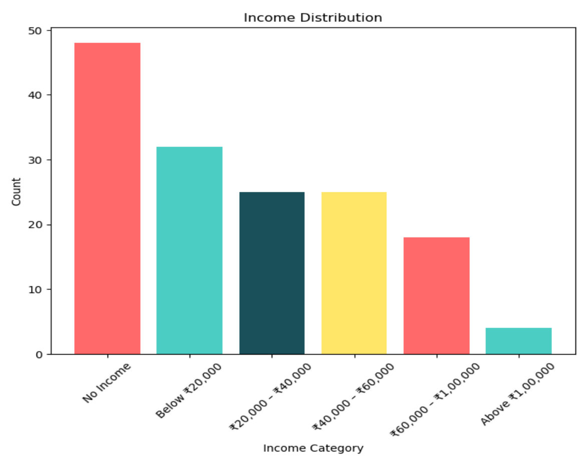
Most responses are concentrated in the mid-range, indicating a large group of undecided users.Extreme low or high intent responses are fewer, showing moderate overall inclination.This reflects a potential opportunity for conversion through targeted strategies.



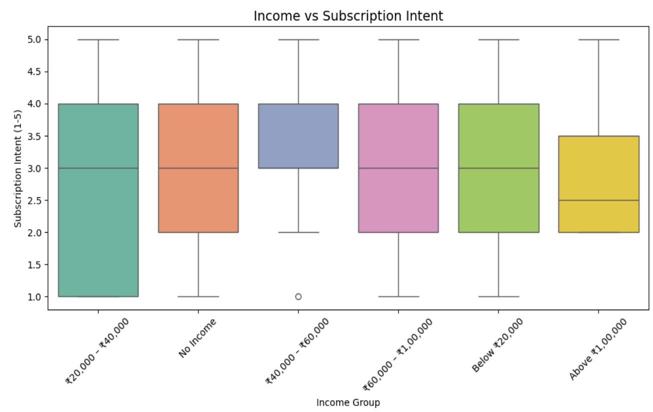
Middle-aged users (especially 35–44) show relatively higher and more consistent subscription intent compared to other groups, indicating stronger willingness to payYounger users exhibit higher variability in intent, suggesting mixed behavior where some are highly interested while others rely on free alternatives. Older users tend to show lower and more stable intent, implying weaker engagement with OTT platforms or lower perceived value.



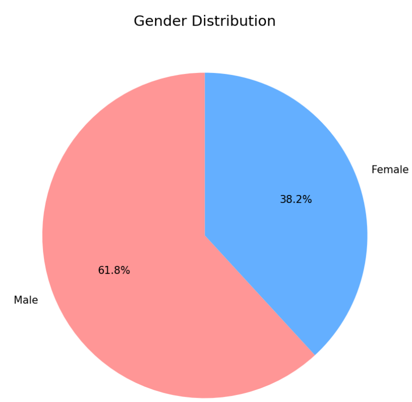
The sample is predominantly composed of younger users, indicating that the analysis mainly reflects the behavior of digitally engaged and tech-savvy consumers.Older age groups are underrepresented, which limits the ability to generalize findings across all demographic segments. The dataset is skewed toward early-stage earners or students, suggesting that financial constraints and value perception may play a key role in subscription decisions.



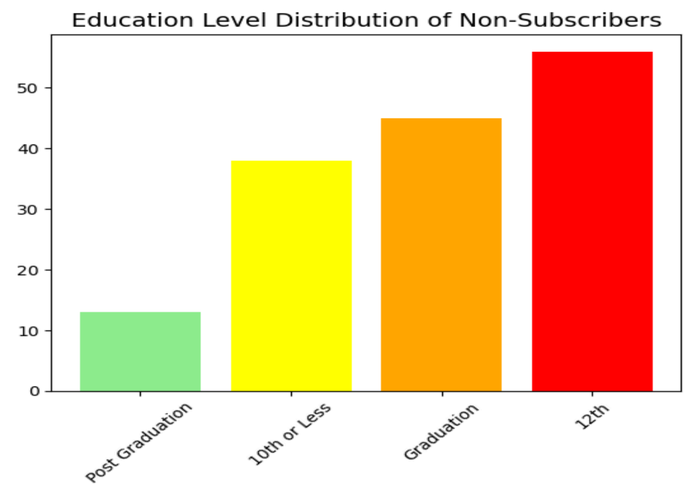
The sample is dominated by no-income and lower-income respondents, so the dataset mainly represents budget-sensitive users.Higher-income groups are small, which makes their pattern less stable for strong conclusions.This means the analysis is largely centered on price-conscious non-subscribers.



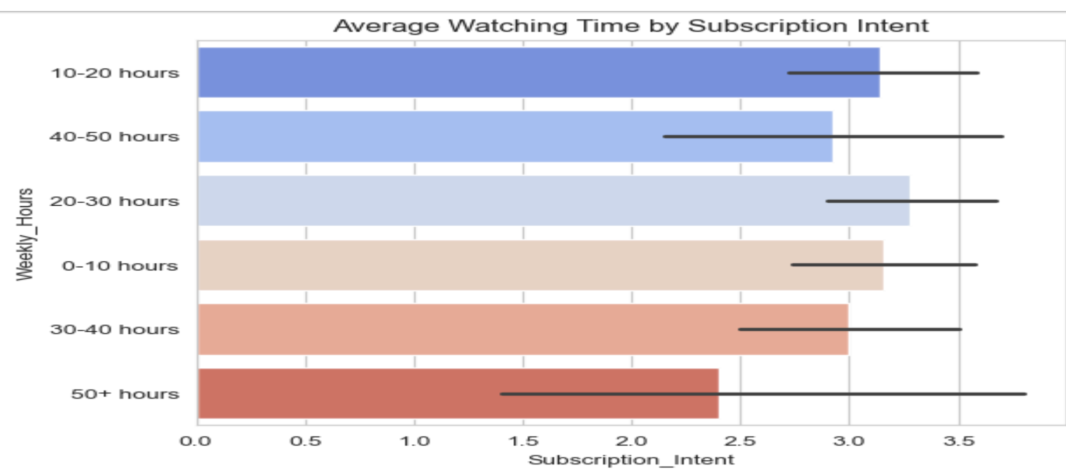
Subscription intent stays broadly similar across most income groups, so income is not a strong differentiator.Mid-income groups appear slightly more positive, but the pattern is not strong enough to suggest a clear trend.Even higher-income users do not show consistently higher intent, which suggests that affordability alone is not the main issue.



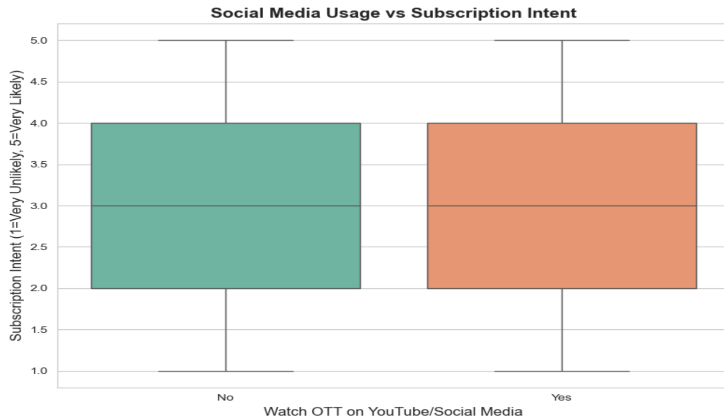
The sample is slightly skewed toward male respondents, indicating a moderate imbalance in representation.Both genders are sufficiently represented to allow general insights, but results may slightly reflect male user behavior more.This suggests findings are broadly applicable but should consider potential gender-based preference differences.



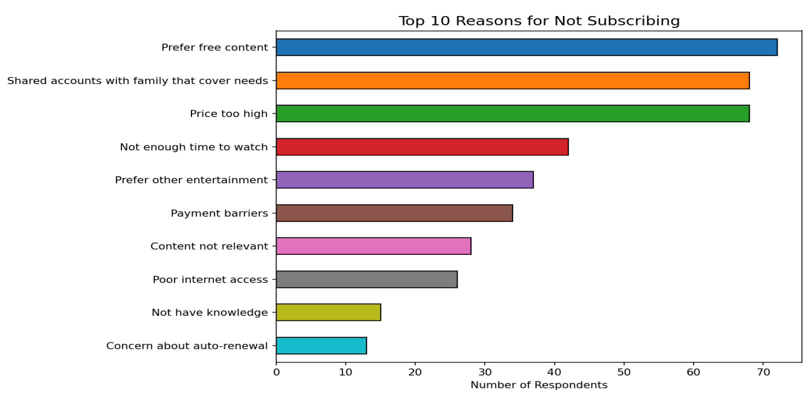
Most respondents are from 12th and graduation levels, so the sample is mainly educated youth and early-degree users.Post-graduate respondents are few, so their behavior should not be overgeneralized. The education mix suggests the study reflects a relatively young and academically active population.



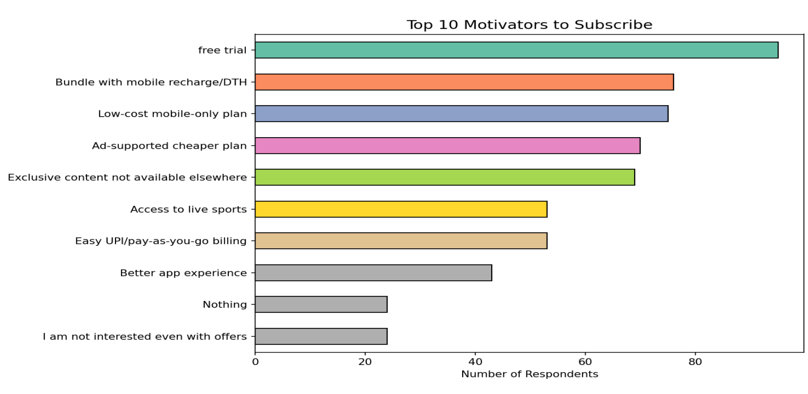
Subscription intent remains relatively consistent across different viewing time groups.Higher watching time does not strongly translate into higher subscription intent.This suggests that engagement level alone is not enough to drive conversion.



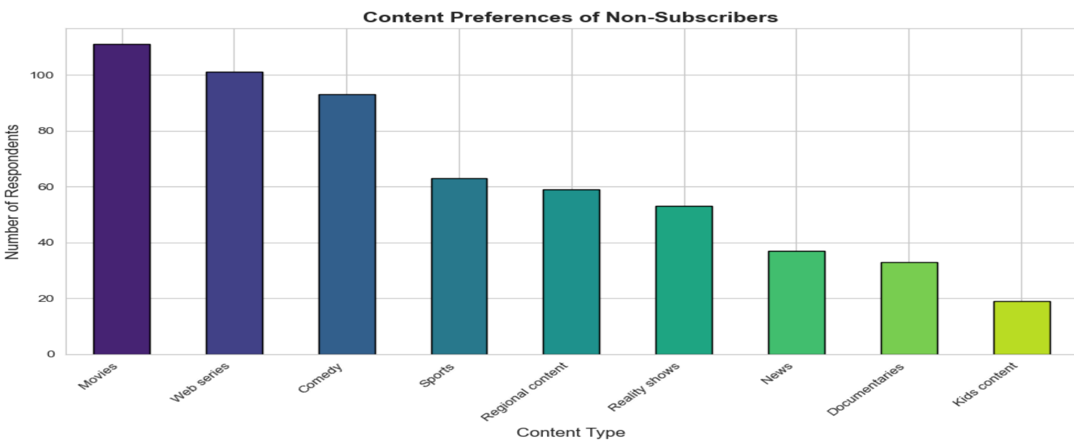
There is minimal difference in subscription intent between users who watch OTT content via social media and those who do not.The overlapping distributions indicate no strong impact of social media usage on intent.This suggests that social platforms are not a key driver of paid OTT adoption in this dataset.



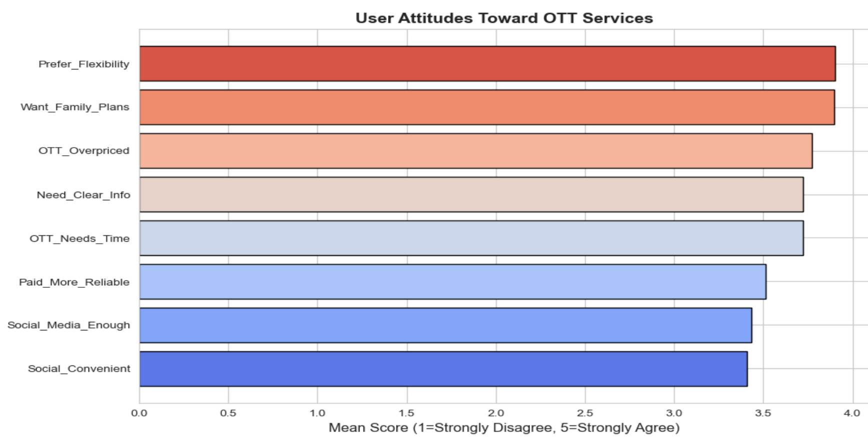
Preference for free content and availability of shared accounts are the primary barriers, indicating strong substitutes exist.Pricing concerns remain a major factor, reinforcing affordability issues.Time constraints and alternative entertainment options also reduce the perceived need for subscriptions.



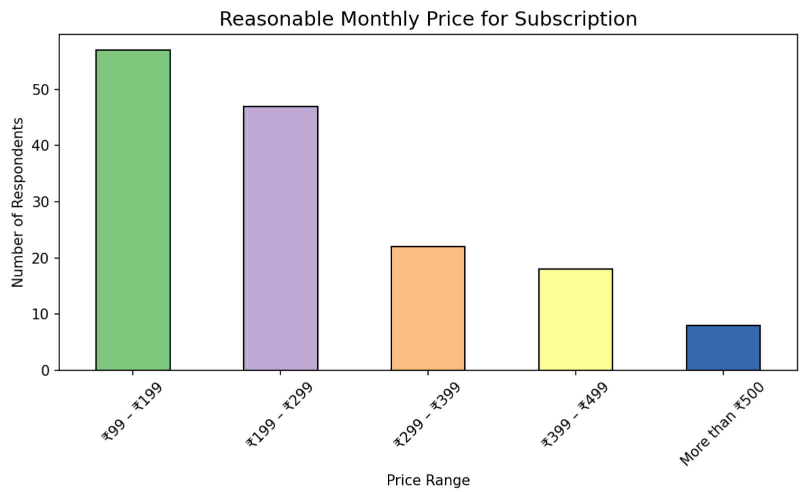
Free trials and bundled offers emerge as the strongest motivators, indicating users prefer low-risk entry options.Cost-related options like cheaper plans and flexible pricing also play a significant role.Exclusive content and added benefits are important but secondary to pricing and accessibility factors.



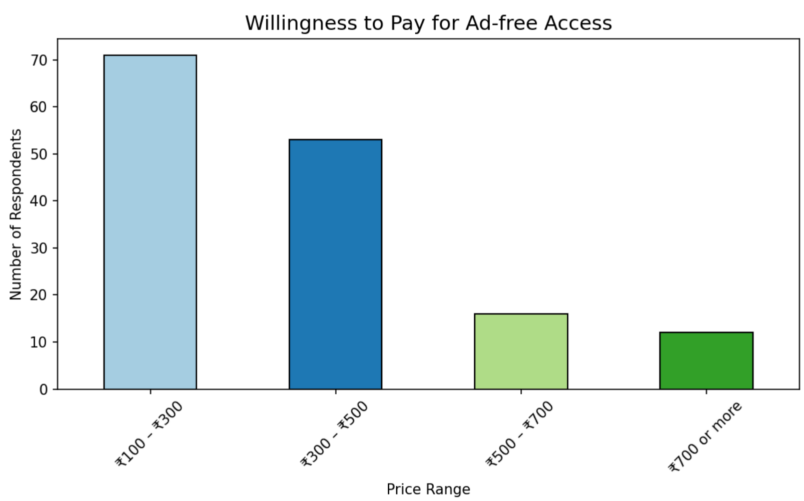
Movies, web series, and comedy are the most preferred content types, indicating strong demand for mainstream entertainment.Niche categories like documentaries, news, and kids content show relatively lower interest among non-subscribers.Sports and regional content have moderate demand, suggesting potential for targeted content strategies.This indicates that content availability alone may not be the issue, as highly demanded content already exists in popular categories.



Users strongly prefer flexible subscription options and family plans, highlighting the importance of convenience and shared value.OTT services are generally perceived as overpriced, indicating a gap between cost and perceived value.There is a clear need for better information and transparency, suggesting users lack clarity about offerings.Time constraints and moderate agreement on reliability indicate that practical usage factors also influence subscription decisions.



Users perceive lower pricing tiers as more acceptable, reinforcing a strong value-conscious mindset.Acceptance declines as prices increase, indicating resistance to higher subscription costs.This highlights a gap between perceived value and current pricing expectations.



Most users prefer lower price ranges, indicating strong price sensitivity among non-subscribers. Interest drops as price increases, showing limited willingness to pay premium amounts.
This suggests that affordability is a key constraint in converting unpaid users.

Feature Engineering

- Raw categorical survey data was transformed into numerical format to make it suitable for statistical analysis and machine learning models. Ordinal variables such as age, income, and usage frequency were mapped based on their natural order to preserve meaningful relationships.
- Binary variables (e.g., Yes/No responses) were converted into 0–1 format to simplify interpretation and enable model compatibility. This allows a clear distinction between presence and absence of behaviors.
- Price-related variables were converted into numeric midpoints to quantify willingness to pay and perceived reasonable price. A new feature, **Price Gap**, was created to capture the difference between perceived value and actual willingness, helping identify value perception patterns.
- Behavioral intensity was captured using aggregated features such as **Motivator Count** and **Barrier Count**, which summarize how many factors influence a user's decision. These help identify whether users are decision-ready or require multiple triggers.
- The target variable was transformed into a binary classification (High vs Low Intent) to align with the business objective of identifying potential subscribers rather than predicting exact intent scores.

Demographic Variables vs Subscription Intent (Non-Parametric Analysis)

Why Non-Parametric Tests Were Used ?
The dependent variable **Subscription Intent** is measured on a **Likert scale (1–5)**, which is ordinal in nature and does not satisfy the assumptions of normality required for parametric tests like ANOVA. Additionally, demographic variables include a mix of **ordinal (Age, Income, Education)** and **nominal (Occupation, Region)** categories.

Therefore, **Kruskal-Wallis test** was used as it:

- Compares median differences across multiple groups
- Does not assume normal distribution
- Is suitable for ordinal data

For each demographic variable:

- **H_0 (Null Hypothesis):**There is no significant difference in subscription intent across different groups of the variable.
- **H_1 (Alternative Hypothesis):**At least one group has a significantly different subscription intent.

Age

- p-value = **0.0487** Significant

Post-hoc Analysis for Age
Dunn's Test with Bonferroni correction was applied
- To identify **which specific age groups differ**
- Kruskal-Wallis only tells "difference exists", not "where"

Result:

- Significant differences observed between:
  - 35–44 vs younger groups
  - 35–44 vs 45–54

Middle-aged users have significantly higher intent compared to others

Multi-Group Variables (Kruskal–Wallis Test + Descriptives)

Variable	p-value	Significant (α=0.05)	Key Group Insight (Mean / Median)
Age	0.0487	Yes	Highest: 35–44 (Mean = 3.83, Median = 4.5) Lowest: 45–54 (Mean = 2.75, Median = 2.5)

Education	0.1387	No	Post Graduation highest (Mean = 3.77, Median = 4) but small sample
Occupation	0.8760	No	Minor variation; most groups around Mean ≈ 3
Region	0.3988	No	Rural slightly higher (Mean ≈ 3.26) vs Urban lower (Mean ≈ 2.91)
Income	0.8226	No	Similar across groups; no clear trend with increasing income

Although postgraduates show higher average intent, the difference is not statistically strong and education alone does not meaningfully influence subscription decisions. Subscription intent is fairly similar across occupations, and extreme values like government employees are due to very small sample size and are not reliable indicators. Location does not play a major role in determining subscription intent, as users from metro, rural, and urban areas behave similarly in their willingness to subscribe. Contrary to expectations, income does not significantly impact subscription intent, and even higher-income users do not show consistently higher willingness to subscribe.

Gender vs Subscription Intent (Mann–Whitney U Test)

Why Mann–Whitney U Test?

Gender is a binary categorical variable (Male/Female) and the dependent variable Subscription Intent is ordinal (Likert scale 1–5).

- It compares distributions between **two independent groups**
- It does not assume normality
- It is suitable for **ordinal data**

H_0 (Null Hypothesis):There is no significant difference in subscription intent between male and female users.

H_1 (Alternative Hypothesis): There is a significant difference in subscription intent between male and female users.

p-value: 0.0407 Significant at **α = 0.05 (**Male: Mean = 3.29, Median= 3 Female: Mean= 2.84, Median= 3)

- Male users appear slightly more inclined toward paid OTT subscriptions compared to female users.
- However, the difference is not large enough to build gender-specific strategies alone.
- Gender may act as a supporting factor, but not a primary driver of subscription decisions.

- Among all demographic variables, only Age and Gender show statistically significant differences in subscription intent, indicating a limited but noticeable influence. Middle-aged users tend to exhibit higher intent, while male users show slightly higher inclination compared to female users.
- However, the mean and median values across all groups remain relatively close, suggesting that these differences are moderate rather than strong. This indicates that demographic factors alone do not create sharp distinctions in user behavior.
- Other variables such as Education, Income, Occupation, and Region are not statistically significant, reinforcing that subscription intent is largely independent of traditional demographic segmentation.
- Overall, the findings suggest that subscription decisions are not primarily driven by who the users are (demographics), but rather by how they behave and what they perceive (behavioral and attitudinal factors).

Behavioral Factors vs Subscription Intent

Methodology

- Binary variables → Mann–Whitney U Test
- Multi-category variable → Kruskal–Wallis Test
- Continuous/ordinal → Spearman Correlation

H_0: No significant difference/relationship between behavioral variable and subscription intent

H_1: Significant difference/relationship exists

Binary Variables (Mann–Whitney U Test)

Variable	p-value	Significant (α=0.05)	Mean (Yes vs No)	Median	High Intent % (Yes vs No)
Has_Cable	0.5659	No	3.08 vs 3.20	3 vs 3	38.89 vs 43.18
Watch_YouTube_Social	0.9247	No	3.12 vs 3.11	3 vs 3	39.13 vs 43.24
Had_Subscription_Before	0.8829	No	3.09 vs 3.13	3 vs 3	43.64 vs 38.14
Use_Free_OTT	0.6532	No	3.06 vs 3.18	3 vs 3	41.77 vs 38.36

- All binary behavioral variables are statistically non-significant.
- Mean and median values are nearly identical across groups, indicating minimal behavioral difference.
- High Intent % varies slightly but does not form a consistent pattern.

Insight: These behavioral factors individually do not strongly influence subscription intent.

Multi-category Variable (Kruskal–Wallis Test)

Variable	Test	p-value	Significant (α=0.05)
Watch_OTT_Freq	Kruskal–Wallis	0.031	Yes

Post-hoc Analysis
Test Used: Dunn’s Test with Bonferroni Correction
Why Dunn’s Test?

- Kruskal–Wallis only tells “difference exists”
- It does NOT tell which groups differ
- Dunn’s test is standard post-hoc for non-parametric multi-group comparison

Why Bonferroni?

- Multiple comparisons → control Type I error
- Makes results statistically reliable

Group Comparison	p-value (Adjusted)	Significant
Shared vs No OTT	< 0.05	Yes
Rarely vs No OTT	< 0.05	Yes
Free vs No OTT	> 0.05	No
Shared vs Free	borderline	No
Rarely vs Free	> 0.05	No
Shared vs Rarely	> 0.05	No

Comparison	Interpretation
Shared accounts vs No OTT users	Significant difference
Rarely vs No OTT users	Significant difference
Shared accounts vs Free content	Mild difference (may or may not be significant depending on correction)

- Users who access OTT via shared accounts show significantly higher subscription intent compared to those who do not watch OTT at all.
- Similarly, even occasional users (rarely) have higher intent than non-users.

Ordinal Variable (Spearman Correlation)

Variable	Test	Correlation (ρ)	p-value	Significant
Weekly_Hours_Num	Spearman	-0.068	0.4020	No

Very weak negative correlation between weekly hours and intent
Not statistically significant

- Most behavioral variables such as **cable usage, social media exposure, prior subscription, and use of free OTT platforms** do not show a statistically significant relationship with subscription intent, indicating that these factors alone are not strong drivers of conversion.
- **Time spent on OTT platforms also does not significantly influence intent**, suggesting that higher engagement or consumption does not necessarily translate into willingness to pay.
- However, **OTT usage frequency shows a significant effect**, where users already exposed to OTT (especially through shared accounts or occasional viewing) demonstrate higher subscription intent compared to non-users.
- Overall, the findings indicate that **mere exposure or usage behavior is not sufficient to drive subscription decisions**; instead, intent is likely influenced by deeper factors such as perceived value, pricing, and content relevance.

Attitudinal Factors vs Subscription Intent

Methodology

- Test Used Spearman Rank Correlation (ρ)

Reason: Because both attitude ratings and subscription intent are ordinal Likert-scale variables (1–5), Spearman's rank correlation was used to measure their monotonic relationship without requiring the assumption of normality.

Hypothesis

H_0: No significant relationship between attitudinal variable and subscription intent
H_1: Significant relationship exists

Variable	ρ (rho)	p-value	Significant (α=0.05)	Interpretation
Social_Media_Enough	-0.028	0.7338	No	No relationship
Social_Convenient	0.082	0.3137	No	Very weak positive
OTT_Needs_Time	-0.015	0.8529	No	No relationship
OTT_Overpriced	-0.106	0.1944	No	Weak negative
Paid_More_Reliable	0.159	0.045	Yes	Weak positive relationship
Prefer_Flexibility	-0.123	0.1319	No	Weak negative
Need_Clear_Info	0.035	0.6688	No	No relationship
Want_Family_Plans	0.063	0.4437	No	Weak positive

Only Paid_More_Reliable shows a statistically significant relationship with subscription intent.
The positive correlation indicates that users who believe paid OTT services are more reliable tend to have higher intent to subscribe.
All other variables show weak or negligible correlations, indicating no strong linear or monotonic relationship.
While higher agreement that paid OTT is more reliable consistently drives greater subscription intent, a stronger preference for flexibility is associated with lower intent – indicating commitment avoidance – and the belief that OTT is overpriced does not strongly differentiate users, suggesting that price concern is a universal sentiment rather than a distinct barrier.

Most attitudinal variables show **no strong statistical relationship**, indicating that attitudes alone do not clearly differentiate users.
However, **perceived value (reliability of paid OTT)** plays a key role in increasing subscription intent.
This suggests that **value perception, not just features or convenience, drives conversion decisions**.

Variable	Level	Mean	Count	High Intent %
Social_Media_Enough	1	3.44	16	62.50
	2	3.06	17	35.29
	3	3.05	37	37.84
	4	3.02	49	34.69
	5	3.21	33	42.42
Social_Convenient	1	2.83	12	25.00
	2	3.08	26	34.62
	3	2.86	37	32.43
	4	3.50	42	54.76
	5	3.06	35	40.00
OTT_Needs_Time	1	3.00	10	40.00
	2	3.00	15	46.67
	3	3.41	27	44.44
	4	3.04	55	34.55
	5	3.11	45	42.22
OTT_Overpriced	1	2.43	7	28.57
	2	3.53	19	52.63
	3	3.19	26	42.31
	4	3.28	50	46.00
	5	2.86	50	30.00

Paid_More_Reliable	1	2.82	11	36.36
	2	2.96	25	28.00
	3	2.88	24	33.33
	4	3.17	59	40.68
	5	3.42	33	54.55
Prefer_Flexibility	1	4.00	5	60.00
	2	3.00	10	40.00
	3	3.17	30	46.67
	4	3.25	57	49.12
	5	2.88	50	24.00
Need_Clear_Info	1	3.22	9	55.56
	2	3.07	14	35.71
	3	3.10	31	41.94
	4	3.00	54	38.89
	5	3.27	44	38.64
Want_Family_Plans	1	2.71	7	28.57
	2	3.00	10	40.00
	3	3.04	28	42.86
	4	3.17	54	40.74
	5	3.19	53	39.62

Motivators vs Subscription Intent (Mann–Whitney U Test)

Methodology

- Test Used: **Mann–Whitney U Test**
- Reason:
  - Motivators are **binary variables (Yes/No)**
  - Target is **ordinal (Likert 1–5)**
  - Non-parametric suitable for distribution comparison

Note on Skipped Variables

- Variables like **Free Trial, Low Cost, Recommendation** were skipped due to **insufficient variation or imbalance in responses**
  (Either too many Yes or No → test not valid)

Variable	p-value	Adjusted p-value	Significant	Effect Size (r)	Mean (Yes vs No)	High Intent % (Yes vs No)
M_Bundle	0.3813	0.7627	No	0.080	3.03 vs 3.21	35.53 vs 44.74
M_Exclusive	0.6671	0.7627	No	0.040	3.06 vs 3.17	42.03 vs 38.55

None of the motivator variables show statistically significant differences in subscription intent.
Effect sizes are very small, indicating negligible practical impact.
Adjusted p-values (Bonferroni) further confirm lack of significance
Users who selected motivators like bundles or exclusive content do not show higher intent compared to others.
In some cases, users selecting motivators actually show lower intent, suggesting hesitation rather than readiness.

Barriers vs Subscription Intent (Mann–Whitney U Test)

Methodology

Since barriers were collected as **multi-select categorical variables**, they were transformed into **binary indicators (selected vs not selected)**. The dependent variable, **subscription intent**, is measured on a Likert scale, so the **Mann–Whitney U test** was used as it is a non-parametric method suitable for comparing two independent groups without assuming normal distribution.

Additional Checks

- Effect size was calculated using **rank-biserial correlation** to measure the strength of the relationship.
- Group comparisons were supported using **mean values and high-intent proportions** to assess practical significance.
- Since multiple barriers were tested simultaneously, **Holm correction** was applied to adjust p-values and control for Type I error.

Variable	p-value	Adjusted p-value	Effect Size (r)	Mean (Yes vs No)	High Intent (Yes vs No)
B_Price	0.8376	1.0000	0.019	3.09 vs 3.14	41.18 vs 39.29
B_Content	0.6421	1.0000	0.055	3.04 vs 3.14	28.57 vs 42.74

- None of the barrier variables show a **statistically significant relationship** with subscription intent.
- Effect sizes are extremely small, indicating **negligible practical impact**.
- Mean values and high-intent proportions remain **very similar across groups**, reinforcing lack of differentiation.

Barriers are commonly cited by all users, regardless of their intent level, meaning they explain why users are currently not subscribed, but not who is likely to subscribe in the futur*e*.

- Across all statistical tests conducted, only a **few variables showed significant relationships** with subscription intent, while the majority of variables were found to be statistically non-significant.
- Among demographic factors, **Age and Gender demonstrated significant differences**, but the effect was moderate, as mean and median values across groups remained relatively close. Other demographic variables such as income, education, occupation, and region showed no significant influence, indicating that subscription intent is largely independent of traditional demographic segmentation.

- In behavioral analysis, most variables including cable usage, social media exposure, prior subscription, and free OTT usage were **not significant**, suggesting that user activity or engagement alone does not drive subscription decisions. However, **OTT usage frequency showed a significant effect**, where users exposed to OTT (especially via shared accounts) exhibited higher intent compared to non-users.
- Attitudinal analysis revealed that **only the belief that paid OTT services are more reliable** has a statistically significant positive relationship with subscription intent. All other attitudinal variables showed weak or negligible correlations, indicating that general opinions and perceptions do not strongly differentiate users.
- Motivators and barriers were found to be **statistically non-significant**, with very small effect sizes. These variables reflect user preferences and current constraints but do not effectively distinguish between high and low intent users.

Cluster Analysis (OTT User Segmentation)

The goal of clustering was to **segment users based on behavior, attitudes, and price perception** to identify distinct groups with similar characteristics and understand which segments have higher subscription potential.

Features Used for Clustering

Behavioral Features

- Watch_OTT_Freq_num → OTT usage frequency
- Has_Cable → Cable subscription (Yes/No)
- Watch_YouTube_Social → Social/YouTube consumption
- Weekly_Hours_Num → Time spent on content
- Had_Subscription_Before → Past subscription experience
- Use_Free_OTT → Usage of free OTT

Represents **user engagement & consumption habits**

Attitudinal Features

- Social_Media_Enough → Social media replaces OTT
- Social_Convenient → Social platforms are convenient
- OTT_Needs_Time → OTT requires time investment
- OTT_Overpriced → Perception of high pricing
- Paid_More_Reliable → Trust in paid OTT
- Prefer_Flexibility → Avoid long-term commitment
- Need_Clear_Info → Need better understanding
- Want_Family_Plans → Interest in shared plans

Represents **user perception & mindset**

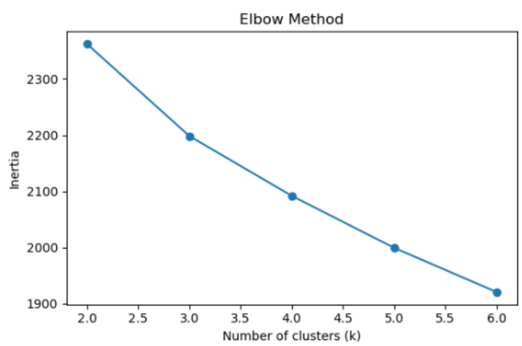
Price & Value Features

- WTP_Num → Willingness to pay
- RP_Num → Expected reasonable price
- Price_Gap → Difference between WTP and expectation

Represents value perception & affordability

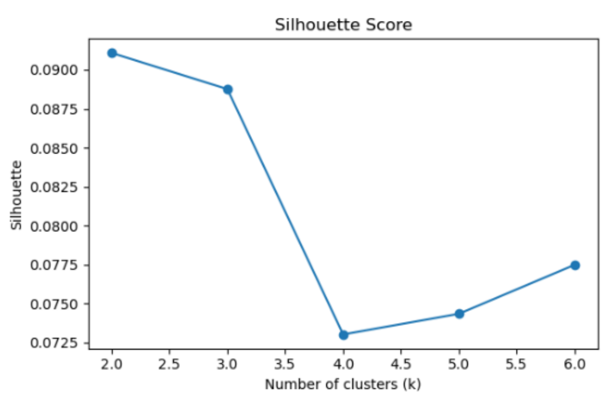
- As the number of clusters (k) increases, inertia consistently decreases, which is expected because more clusters naturally reduce within-cluster variance. However, this alone cannot determine the optimal number of clusters, as it always improves with higher k.

- The silhouette score is highest at k = 2 (0.091) and remains almost similar at k = 3 (0.089), indicating that both configurations produce well-separated and cohesive clusters. However, beyond k = 3, the silhouette score drops significantly, suggesting weaker cluster structure and reduced separation.
- The Davies–Bouldin index decreases as k increases, indicating better separation between clusters. However, this improvement comes at the cost of interpretability, as higher k leads to fragmented and less meaningful segments



- The inertia decreases rapidly from **k = 2 to k = 3**, and then continues to decrease at a slower, more gradual rate.
- This change in slope suggests a **bend (elbow) around k = 3**, indicating that adding more clusters beyond this point does not significantly improve cluster compactness.

Interpretation:The elbow point at k = 3 suggests that three clusters capture most of the structure in the data without unnecessary complexity.



- The silhouette score is highest at **k = 2 (0.091)** and slightly decreases at **k = 3 (0.089)**.
- After k = 3, the silhouette score drops significantly, indicating **weaker cluster separation and increased overlap**.

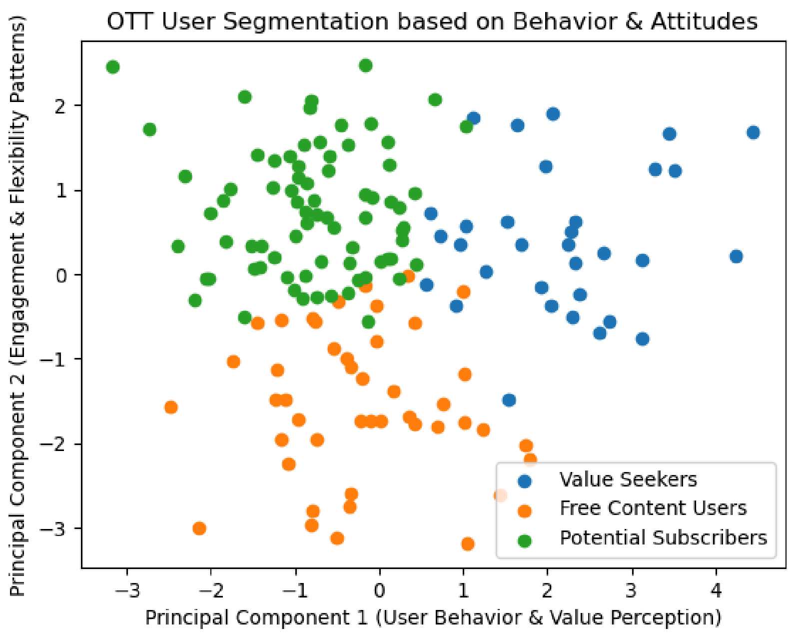
Interpretation: While k = 2 provides the best separation, k = 3 still maintains a strong silhouette score with minimal loss in cluster quality.

A higher silhouette score indicates **better-defined and more distinct clusters**.

k = 3

- Very similar silhouette score to k = 2
- Clear elbow point in inertia
- Provides **more meaningful and interpretable segments**

Cluster Segments (From PCA Plot)



PCA was used to project high-dimensional data into 2D space while retaining maximum information for visualization.”

Cluster 0 (Blue) — Value Seekers

Moderate OTT usage but not fully committed.Higher focus on value and price comparison Slightly better willingness to pay than free users Mixed engagement levels across users.Need strong value proposition Respond well to bundles, discounts, and pricing strategies Focus: *Convince them it’s worth paying*

Cluster 1 (Orange) — Free Content Users

Prefer free OTT or shared content.Lower willingness to pay.Higher preference for flexibility (avoid commitment)Lower engagement with paid ecosystemLowest conversion probability Prefer free or flexible options Focus: *Awareness + freemium, not aggressive conversion*

Cluster 2 (Green) — Potential Subscribers

Higher OTT engagement and usage Strong belief that paid OTT is reliable Higher subscription intent More open to paid conversion**.**Highest conversion potential Already convinced about value Focus: Immediate conversion (offers, trials, urgency)

- The clustering analysis successfully segmented users into **three distinct groups** based on a combination of behavioral patterns, attitudinal perceptions, and price sensitivity. This indicates that users are not homogeneous and exhibit clearly different profiles in terms of engagement and decision-making.
- The clusters highlight that **subscription intent varies more across user segments than across demographic groups**, reinforcing the finding that behavior and perception are stronger drivers than static characteristics.
- One segment consists of **high-potential users who are already engaged and open to subscription**, another represents **value-conscious users who require justification before paying**, and the third includes **free-oriented users with low willingness to commit**.
- The clustering results demonstrate that **exposure, perceived value, and flexibility preferences collectively shape user segmentation**, rather than any single factor alone.

Modelling the Subscription likelihood

The objective of the modeling phase was to predict whether a user is likely to subscribe (High Intent) or not (Low Intent). Since the business goal is to identify potential subscribers, the focus was placed on correctly identifying the positive class.

Target Variable (y)

The target variable used for modeling is:

High_Intent (Binary Classification)

- 1 → High likelihood to subscribe
- 0 → Low likelihood to subscribe

This variable was derived from the original Subscription_Intent (Likert scale) by converting it into a binary outcome to align with the business objective of identifying potential subscribers.

Feature Variables (X)

Behavioral Features

- Watch_OTT_Freq_num
- Has_Cable
- Watch_YouTube_Social
- Weekly_Hours_Num
- Had_Subscription_Before
- Use_Free_OTT

Attitudinal Features

- Social_Media_Enough
- Social_Convenient
- OTT_Needs_Time
- OTT_Overpriced
- Paid_More_Reliable
- Prefer_Flexibility
- Need_Clear_Info
- Want_Family_Plans

Price & Value Features

- WTP_Num (Willingness to Pay)

- RP_Num (Reasonable Price)
- Price_Gap (Perceived value difference)

Model-wise Performance Analysis

1. Logistic Regression

- Accuracy: 0.48
- Recall (Class 1): 0.42

Logistic Regression showed poor overall performance with low accuracy and moderate recall. As a linear model, it assumes a linear relationship between features and the target, which is not suitable in this case where user behavior is complex and non-linear. It struggled to capture underlying patterns in the data.

2. Random Forest

- Accuracy: 0.48
- Recall (Class 1): 0.33

Random Forest did not perform well despite being an ensemble method. The low recall indicates that the model failed to correctly identify potential subscribers. This suggests that the dataset may not have strong feature splits or sufficient signal for tree-based partitioning.

3. Gradient Boosting

- Accuracy: 0.55
- Recall (Class 1): 0.33

Both boosting models showed slight improvement in accuracy compared to Logistic Regression and Random Forest, but recall remained low. These models tend to focus on difficult cases, but in this dataset, they were unable to significantly improve detection of high-intent users.

5. K-Nearest Neighbors (KNN)

- Accuracy: 0.58
- Recall (Class 1): 0.42
- ROC-AUC: 0.669

KNN showed balanced performance across metrics and achieved the highest ROC-AUC among all individual models. This indicates that it has better ranking ability, even if classification performance is moderate. It captures local patterns effectively but lacks generalization.

6. XGBoost

- Accuracy: 0.61
- Recall (Class 1): 0.50
- ROC-AUC: 0.601

XGBoost provided the best balance among individual models, with improved recall and stable accuracy. It captures non-linear relationships and interactions between variables, making it more suitable for this dataset.

Why Recall is Important

In this problem, the goal is to identify users who are likely to subscribe.

- False Negative = Missing a potential subscriber
- False Positive = Targeting a non-subscriber

From a business perspective:Missing a potential subscriber is more costly than targeting a non-subscriber Therefore Recall for the positive class (High Intent) is prioritized over accuracy.

Stacking Model (Final Model)

A stacking ensemble was built using:

- Base Models: SVM, KNN, XGBoost
- Meta Model: Logistic Regression

The idea is to combine strengths of different models:

- SVM → good boundary separation
- KNN → captures local patterns
- XGBoost → captures non-linear relationships

Stacking Performance

- Accuracy: 0.645
- Precision: 0.533
- Recall: 0.667
- F1-score: 0.593
- ROC-AUC: 0.553

```
STACKING MODEL
Accuracy : 0.645
Precision: 0.533
Recall   : 0.667
F1-score : 0.593
ROC-AUC  : 0.553
              precision    recall  f1-score   support

           0       0.75      0.63      0.69        19
           1       0.53      0.67      0.59        12

    accuracy                           0.65        31
   macro avg       0.64      0.65      0.64        31
weighted avg       0.67      0.65      0.65        31
```

Interpretation

- Stacking achieved the **highest recall (0.667)** among all models
- This indicates that the model is able to **identify more potential subscribers**
- Accuracy also improved compared to most individual models
- Although ROC-AUC is moderate, the improvement in recall aligns with the business goal
- Individual models showed **trade-offs between accuracy and recall**
- No single model was able to perform optimally across all metrics
- Stacking combines multiple perspectives and improves overall performance

"The stacking model was selected as the final model because it significantly improves recall, ensuring better identification of potential subscribers, which is the primary business objective."

References

Wasserstein, R. L., & Lazar, N. A. (2016). The ASA's statement on p-values: Context, process, and purpose. *The American Statistician, 70*(2), 129–133.

Cumming, G. (2014). The new statistics: Why and how. *Psychological Science, 25*(1), 7–29.

Neslin, S. A., Gupta, S., Kamakura, W., Lu, J., & Mason, C. H. (2006). Defection detection: Measuring and understanding the predictive accuracy of customer churn models. *Journal of Marketing Research, 43*(2), 204–211.

Cohen, J. (1988). *Statistical power analysis for the behavioral sciences* (2nd ed.). Lawrence Erlbaum Associates.

Rousseeuw, P. J. (1987). Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics, 20*, 53–65.

Reichheld, F. F., & Sasser, W. E. (1990). Zero defections: Quality comes to services. Harvard Business Review, 68(5), 105–111.

Gupta, S., Lehmann, D. R., & Stuart, J. A. (2004). Valuing customers. Journal of Marketing Research, 41(1), 7–18.

Verbeke, W., Martens, D., Baesens, B., & Huysmans, J. (2012). New insights into churn prediction in the telecommunication sector: A profit-driven data mining approach. European Journal of Operational Research, 218(1), 211–229.

Wedel, M., & Kamakura, W. A. (2000). Market segmentation: Conceptual and methodological foundations (2nd ed.). Springer.

Hair, J. F., Black, W. C., Babin, B. J., & Anderson, R. E. (2019). Multivariate data analysis (8th ed.). Cengage Learning.